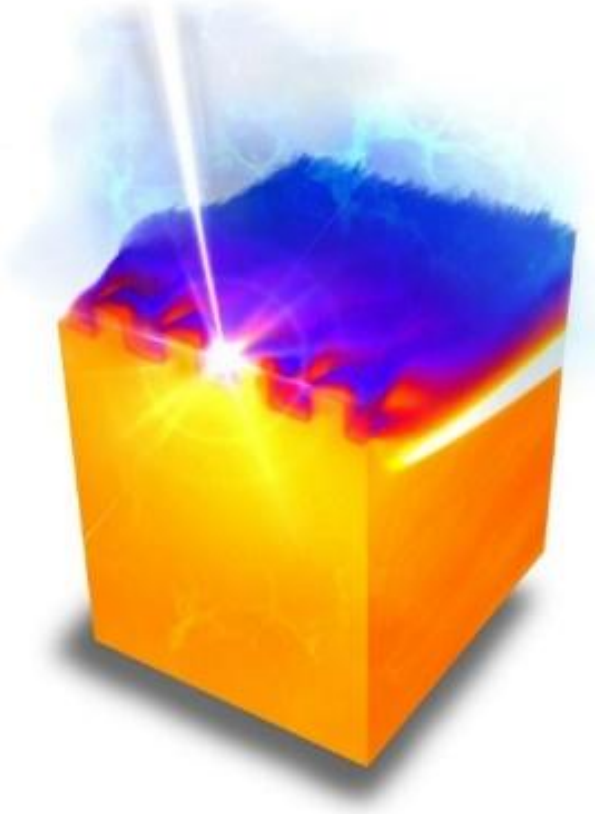
TECHNISCHE UNIVERSITÄT DRESDEN

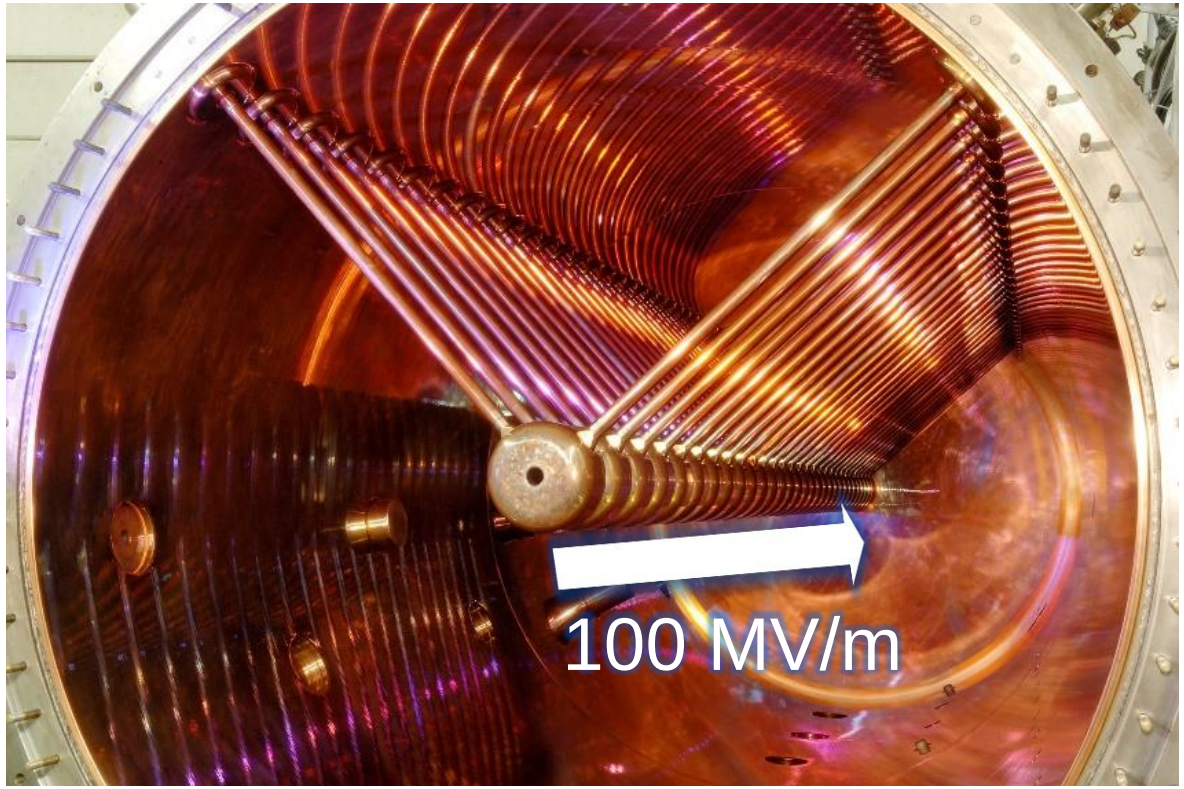
HZDR HELMHOLTZ ZENTRUM DRESDEN ROSSENDORF

Probing of Ultra-Intense Laser-Solid Interaction Dynamics with Resonant Small-Angle X-ray Scattering

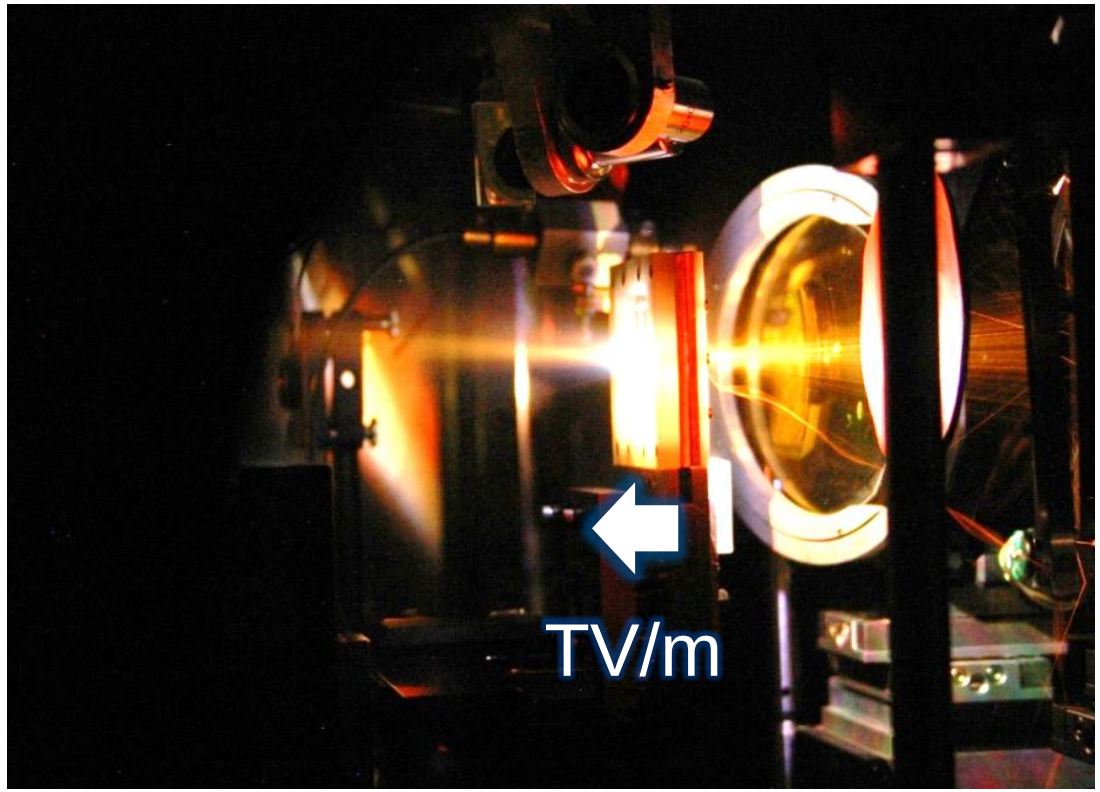
Disputationsvortrag – Lennart Gaus



An alternative concept: Laser-based ion acceleration

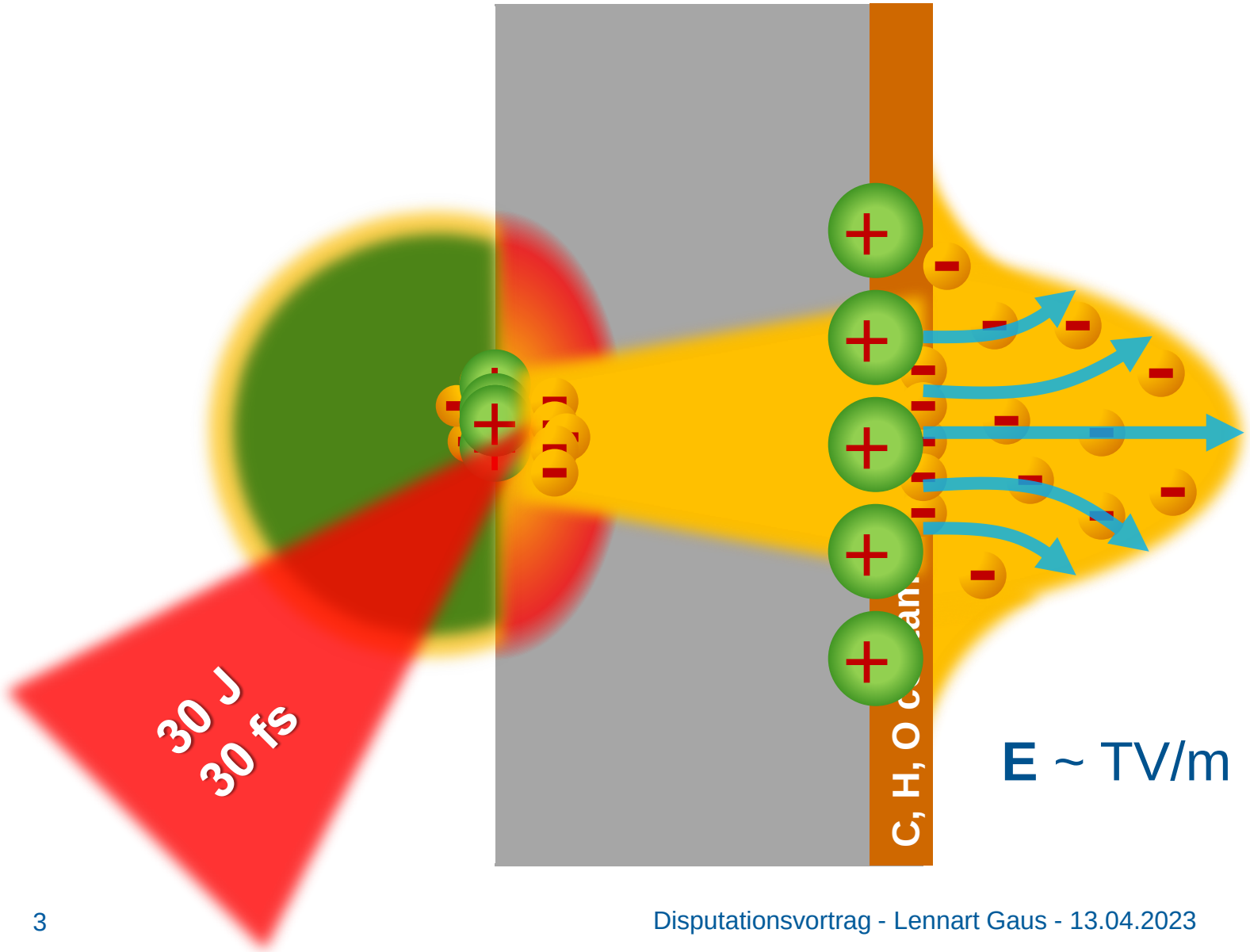


UNILAC Accelerator at GSI

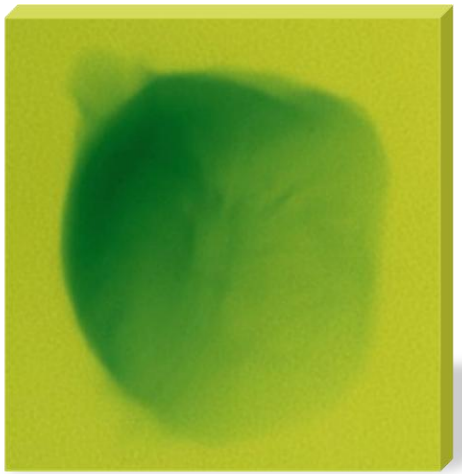


Laser-Ion Acceleration at HZDR

Relativistic Laser-Solid Interaction



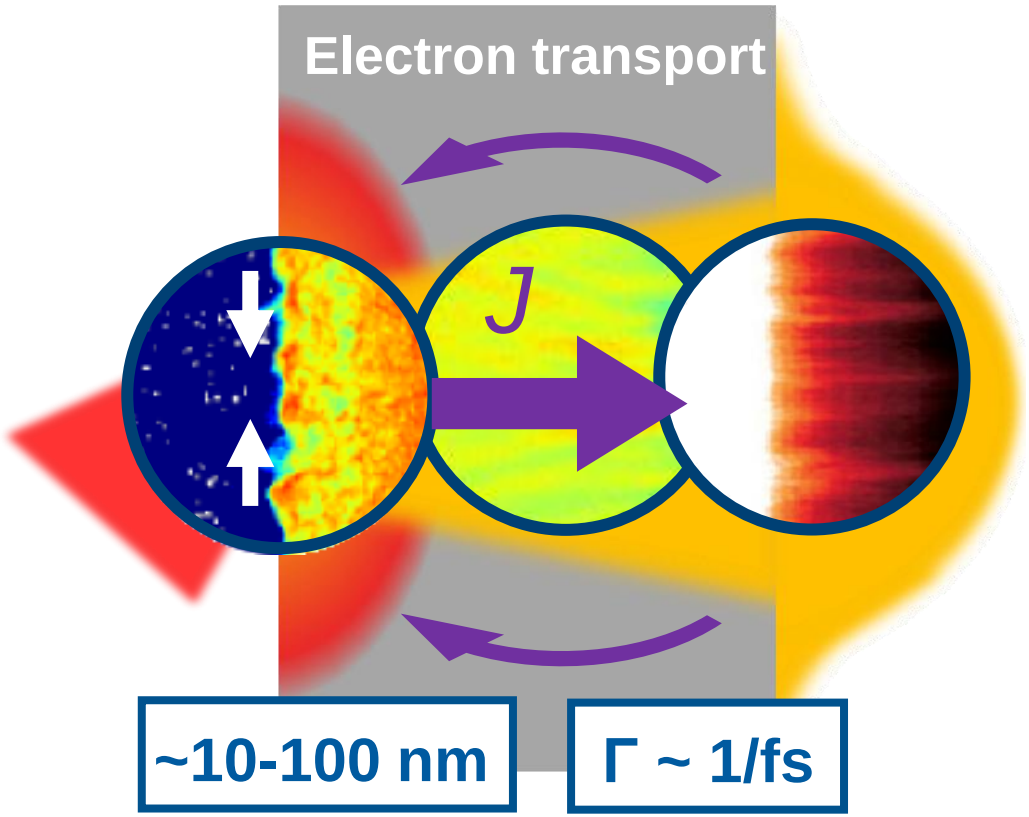
E ~ TV/m

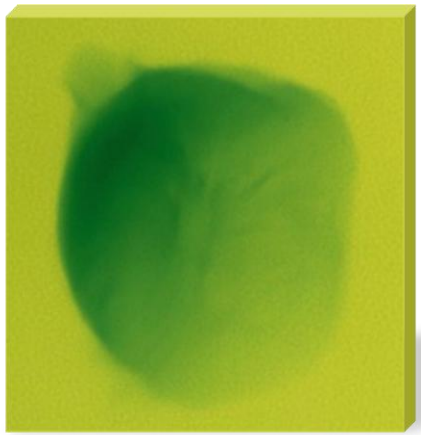


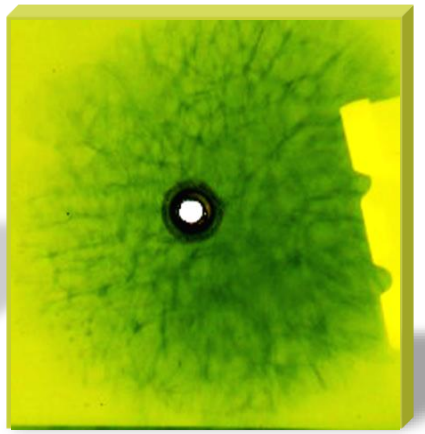
Proton beam spot
(RCF measurement)

Relativistic Laser-Solid Interaction

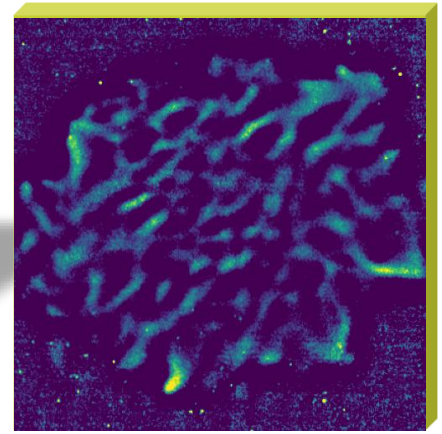
… causes plasma dynamics on the **nm** and **fs scales**







Metzkes-Ng. et al

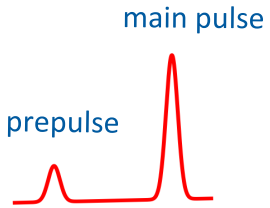


During my PhD

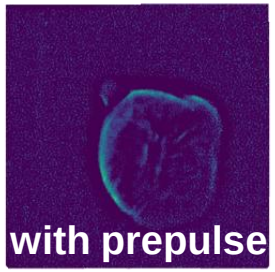
0 ~1 µm few **cm**

Study of p^+ beam modulations in different laser-target settings

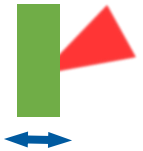
- Temporal laser profile

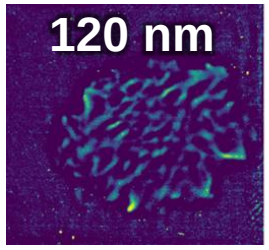


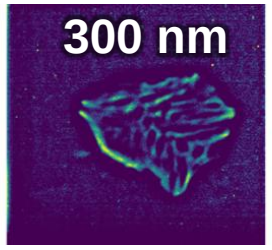


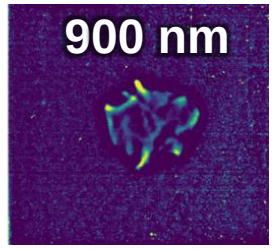


- Target thickness scan



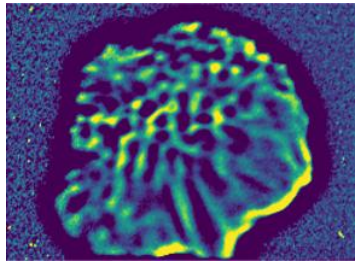




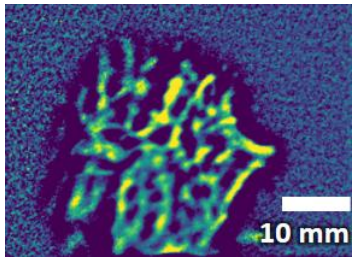


- Laser pulse energy scan

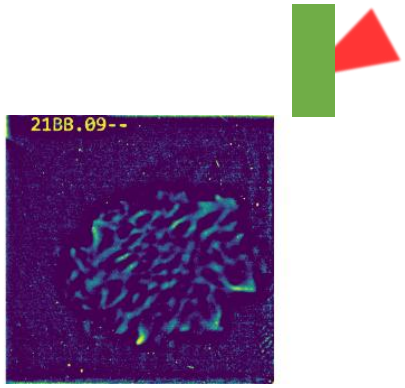
E_L = 18 J

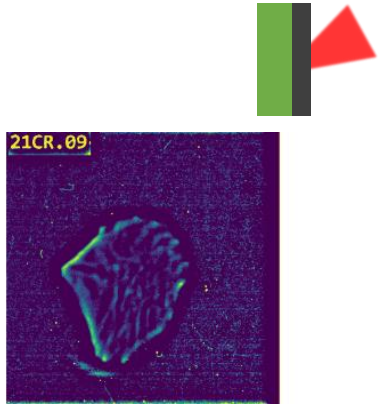


E_L = 5 J

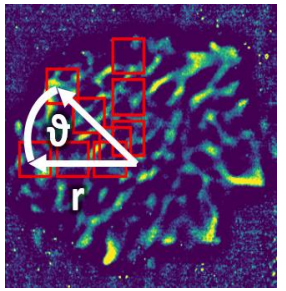


- Target composition



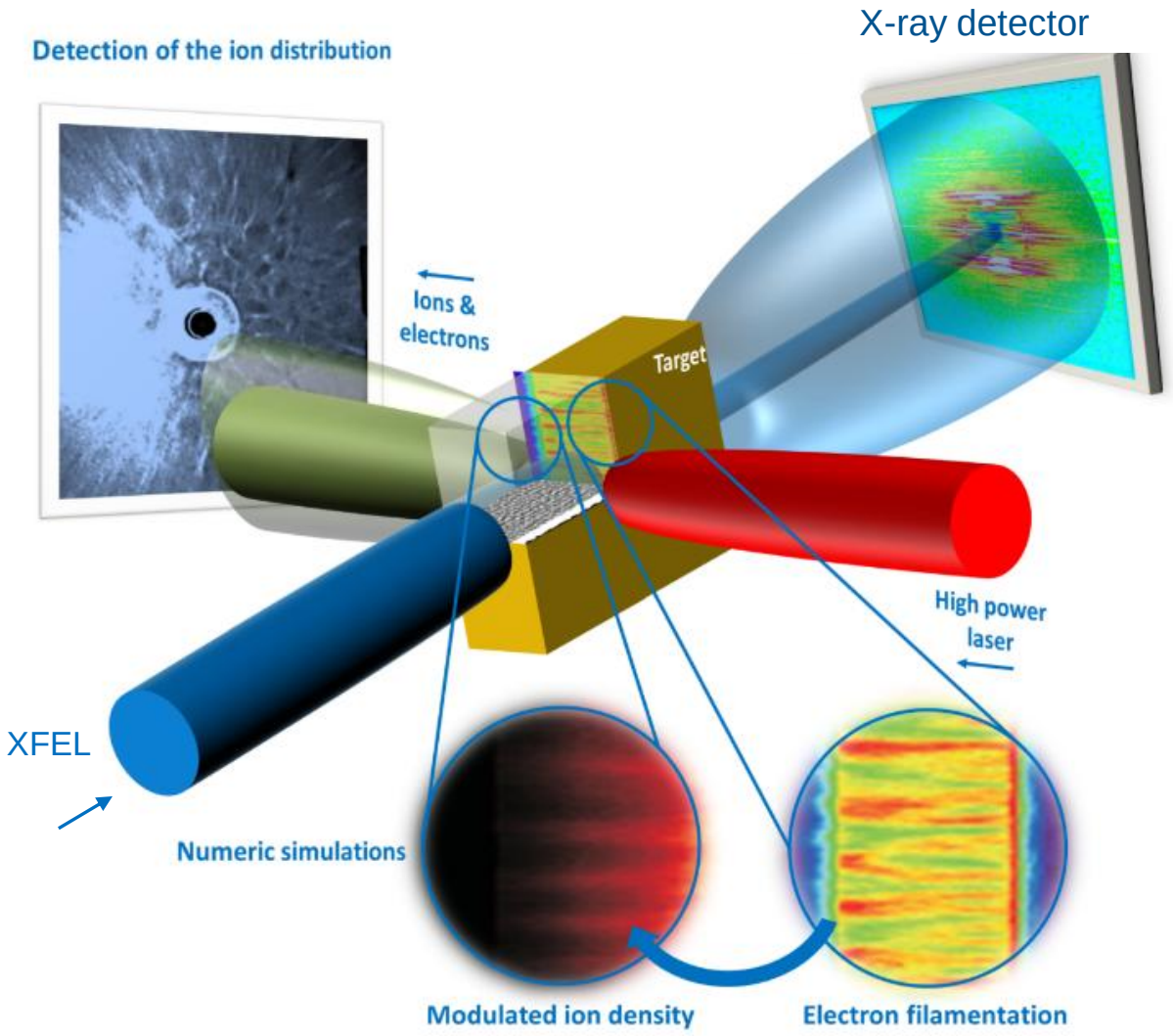


correlation with experimental data and literature



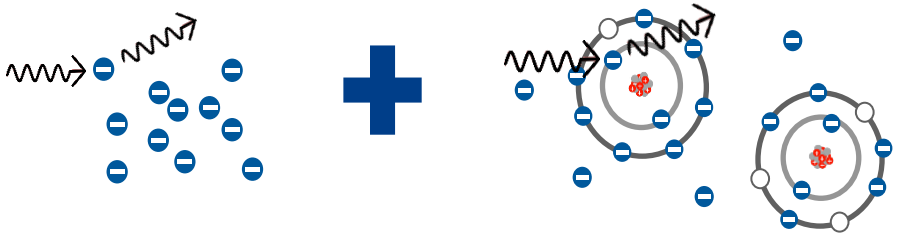
parametrization of the modulation strength

Pump-probe experiments with XFELs are exactly what we



$$I(\vec{q}) \propto |FT\{n_e(\vec{r})\}|^2$$

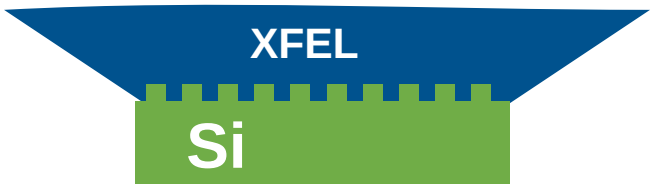
Small-angle X-ray scattering (SAXS)
predominantly by Thomson scattering on:



free electrons bound electrons in ions

T. Kluge et al., Phys. Plasmas **23**, 33103, 2016
T. Kluge et al., Phys. Plasmas **24**,102709, 2017

Methods: Nanostructured gratings as an experimental platform



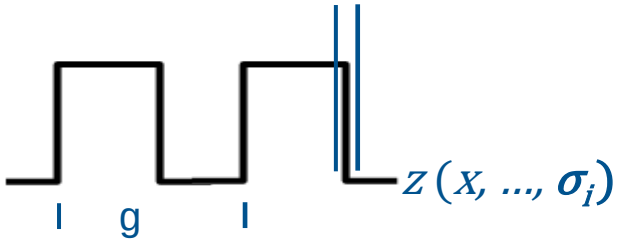
σ_i… *edge smoothness*

$$z \sim \mathrm{erf}\left(\frac{x - x_0}{\sqrt{2}\sigma_i}\right)$$

initial grating

→ XFEL-only

(preshot)



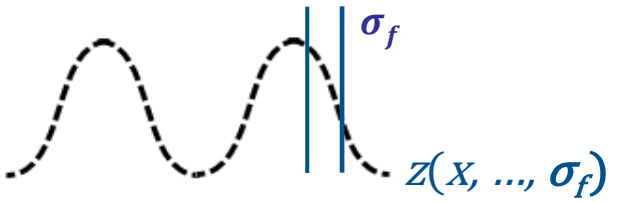
expansion

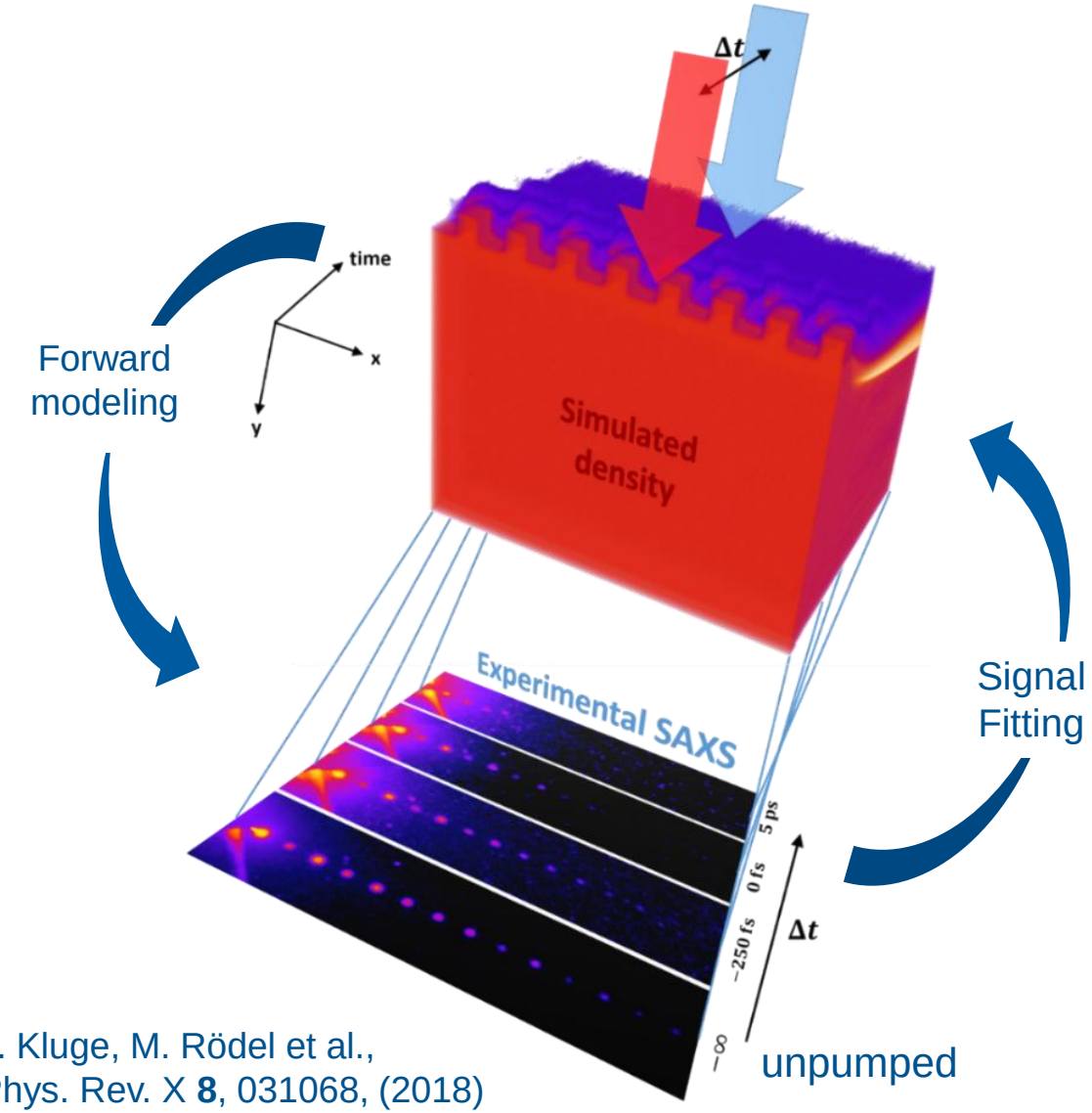
pumped grating

→ HI pump +

XFEL probe

(main shot)





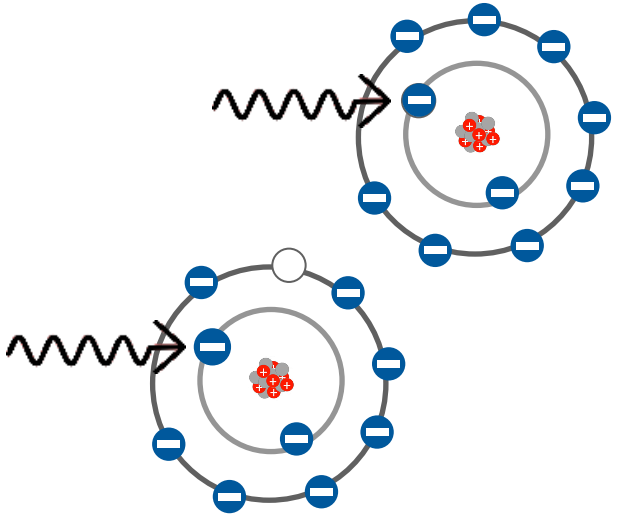
T. Kluge, M. Rödel et al.,
Phys. Rev. X **8**, 031068, (2018)

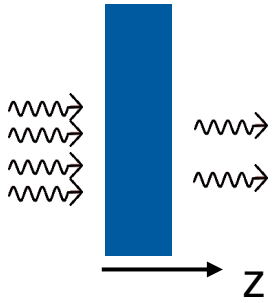
Methods: SAXS signal composition

Absorption processes:

- bound-free transitions (photoionization)

- bound-bound transitions





$$I(z) \sim e^{-\kappa z}$$

κ … attenuation coefficient

$$I(\vec{q}) \propto \left| FT\left\{ n_{e,free}(\vec{r}) + \sum_Q f_Q \cdot n_i(\vec{r}) \right\} \right|^2$$

Scattering on free electrons → **real part**

Absorption in ions → **imaginary part**

Complex ion form factor: $f_Q = f_0 + f' + i \cdot f''$

$\approx Q$ **Absorption**

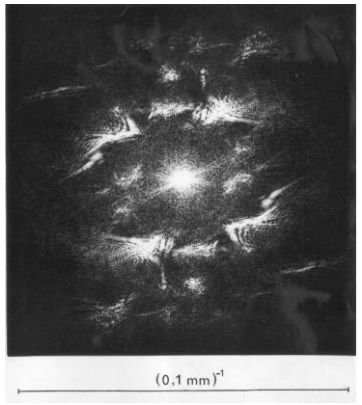
Methods: Symmetry considerations for scattering patterns

Friedel's law: $I(+\vec{q}) = I(-\vec{q})$, if $f(\vec{x})$ is real-valued



real function $f(\vec{x})$

$$I(\vec{q}) = |FT\{f\}|^2$$



central-symmetric diffraction pattern

[1] Friedel G.: Sur les symétries cristallines que peut révéler la diffraction des rayons Röntgen. *Comptes Rendus Acad. Sci.* **157** (1913), 1533–1536.

f … real-valued function $f = f_{\text{even}} + f_{\text{odd}}$ $FT\{f\} = \tilde{f}$

$$FT\{f\} = FT\{f_{even}\} + FT\{f_{odd}\}$$
$$\vdots$$
$$|FT\{f\}|^2 = (\widetilde{f_{even}})^2 + (\widetilde{f_{odd}})^2$$

f* = *S* + *i T is complex

$$FT\{f\} = (FT\{S_{\text{even}}\} + FT\{S_{\text{odd}}\}) + i\,(FT\{T_{\text{even}}\} + FT\{T_{\text{odd}}\})$$
$$\vdots$$
$$|FT\{f\}|^2 = \ldots - 2\widetilde{S_{\text{even}}}\widetilde{T_{\text{odd}}} + 2\widetilde{T_{\text{even}}}\widetilde{S_{\text{odd}}} + \ldots$$

→ **Asymmetry,** if even and odd functions **mix**

f* = *S* + *i T is complex, with ***T* = *α S***

$$FT\{f\} = (1 + i\alpha)(FT\{S_{even}\} + FT\{S_{odd}\})$$
$$\vdots$$
$$|FT\{f\}|^2 = (1 + \alpha^2)\left(\widetilde{S_{even}}^2 + \widetilde{S_{odd}}^2\right)$$

→ **Symmetry is preserved** as even and odd functions **don't** mix

One material: $I(q_y) \sim |(FT\{z_1(x) \cdot n_i(1 + \mathbf{i}\alpha)\}|^2$ with $\alpha = \frac{f''}{f^0 + f'}$

→ We need a **second material** to generate asymmetric scattering patterns

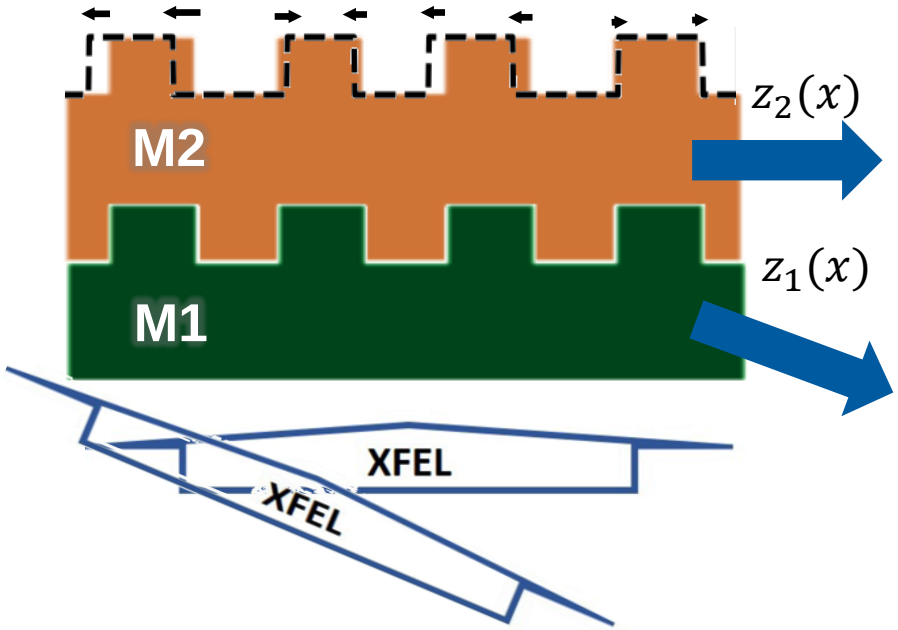
Methods: Symmetry considerations for scattering patterns

Necessary condition to generate asymmetric scattering patterns?

- f = S + iT is complex and T ≠ α S

Sufficient condition? Symmetry break!

- Small shift (imperfect target adjustment)
- Material imperfections



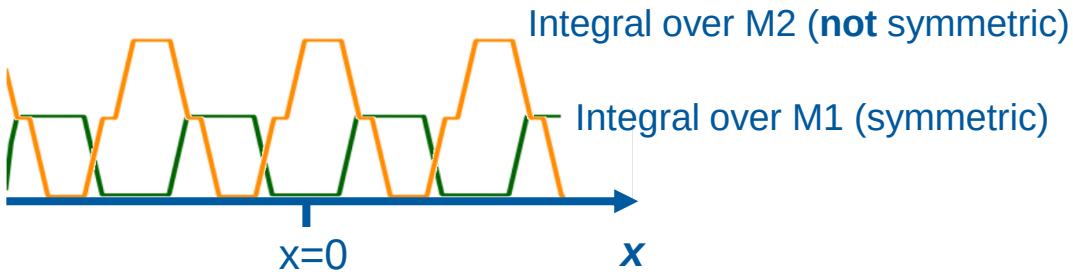
$$I(q_y) \sim |(FT\{I_1\} + FT\{I_2\}|^2$$

$$I_2 \sim (z_2(x) - z_1(x)) \cdot n_{i,M2}(1 + \mathbf{i}\beta)$$

$$\beta = \frac{f''_{M2}}{f^0_{M2} + f'_{M2}}$$

$$I_1 \sim z_1(x) \cdot n_{i,M1}(1 + \mathbf{i}\alpha)$$

$$\alpha = \frac{f''_{M1}}{f^0_{M1} + f'_{M1}}$$

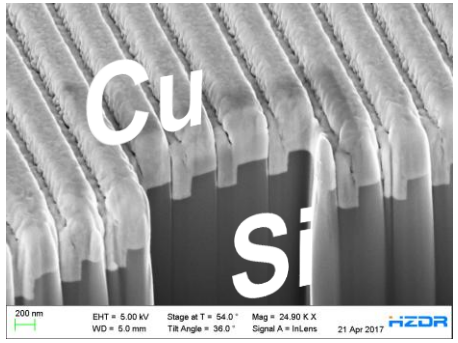


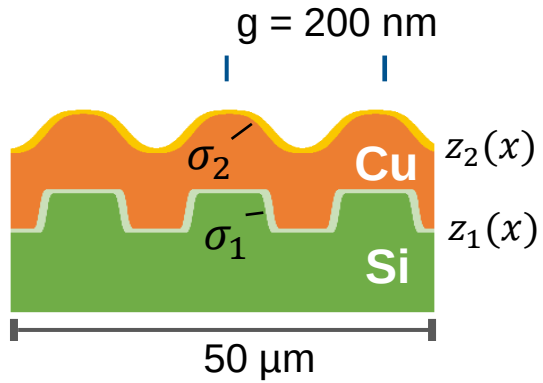
SAXS experiments at MEC



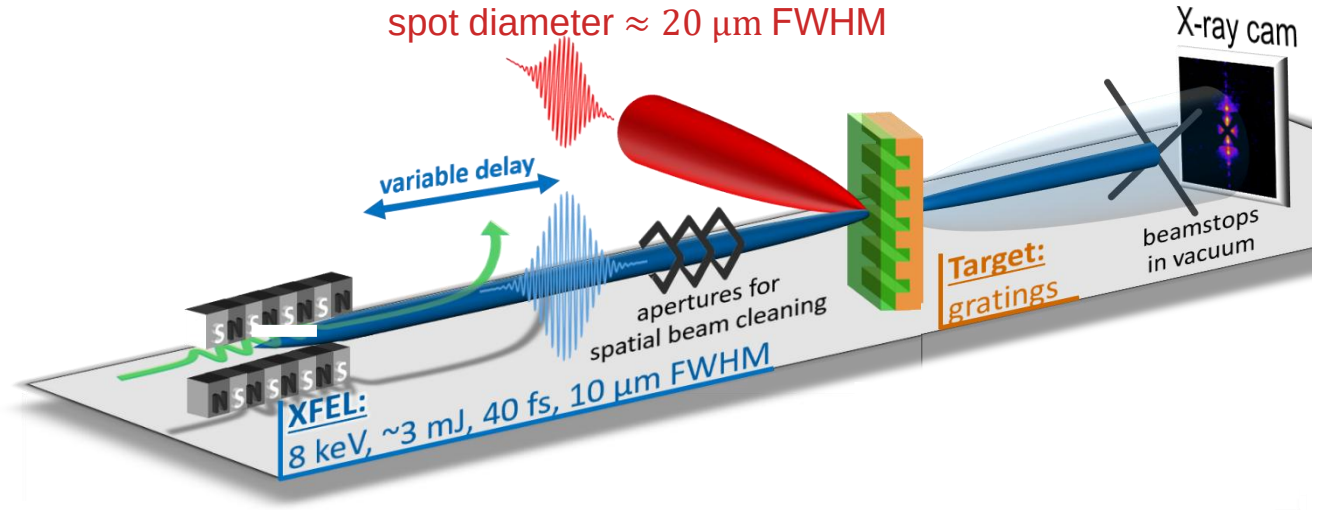
Linac Coherent Light Source at Stanford Linear Accelerator Center

Si-Cu targets





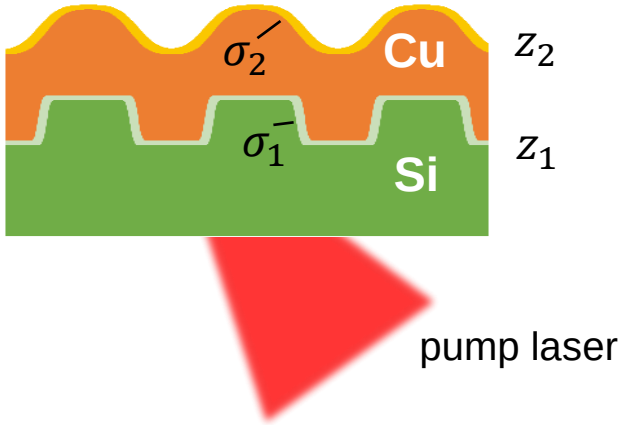
MEC pump laser:
80 fs duration, $I_L \sim 5 \cdot 10^{17}\,\mathrm{W/cm^2}$
spot diameter ≈ 20 µm FWHM



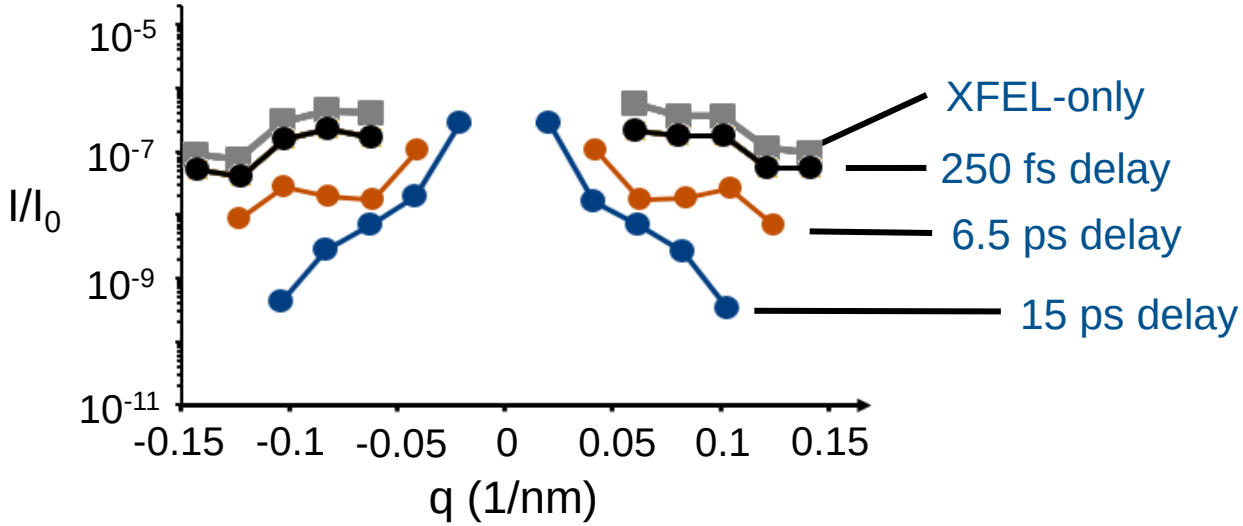
L. Gaus et *al.*, Phys. Rev. Research **3,** 043194, **2021**

My thesis: asymmetry analysis and interpretation

Results: Buried layer expansion



Experimental SAXS data

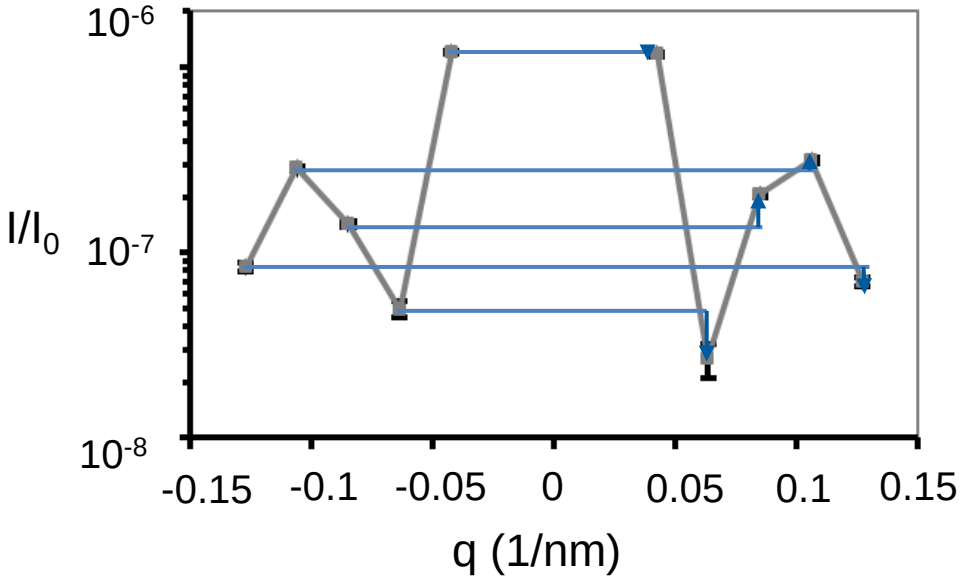


- Grating expansion **on the ps time scale**
- **…** but discrimination between buried and rear grating expansion is not trivial

Results: Asymmetry Analysis

- Asymmetric scattering patterns measured in the unpumped preshots!

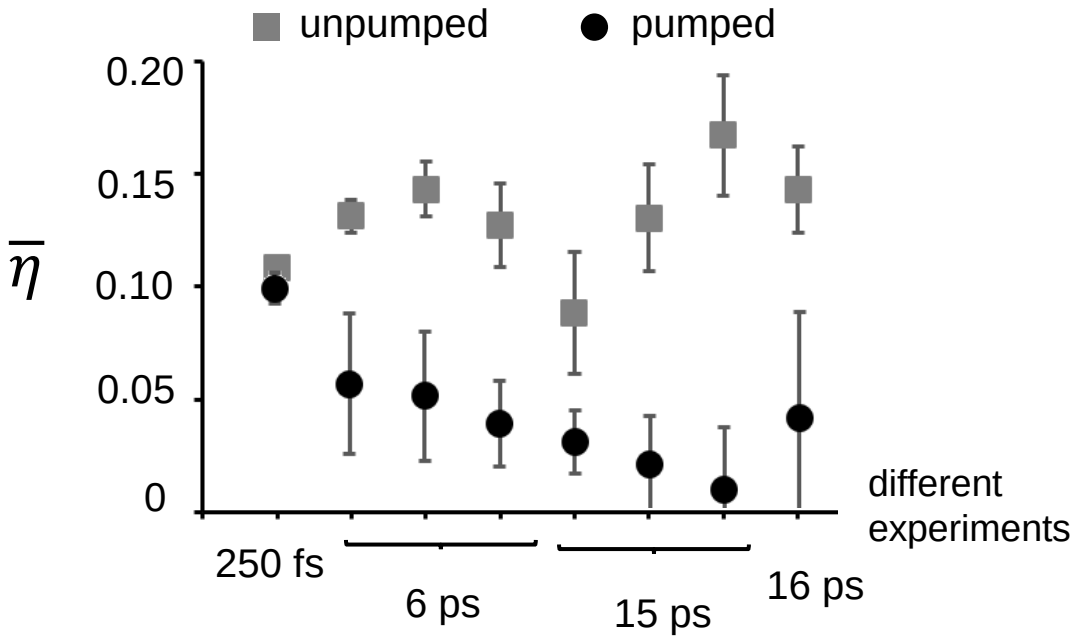
Example of an asymmetric preshot

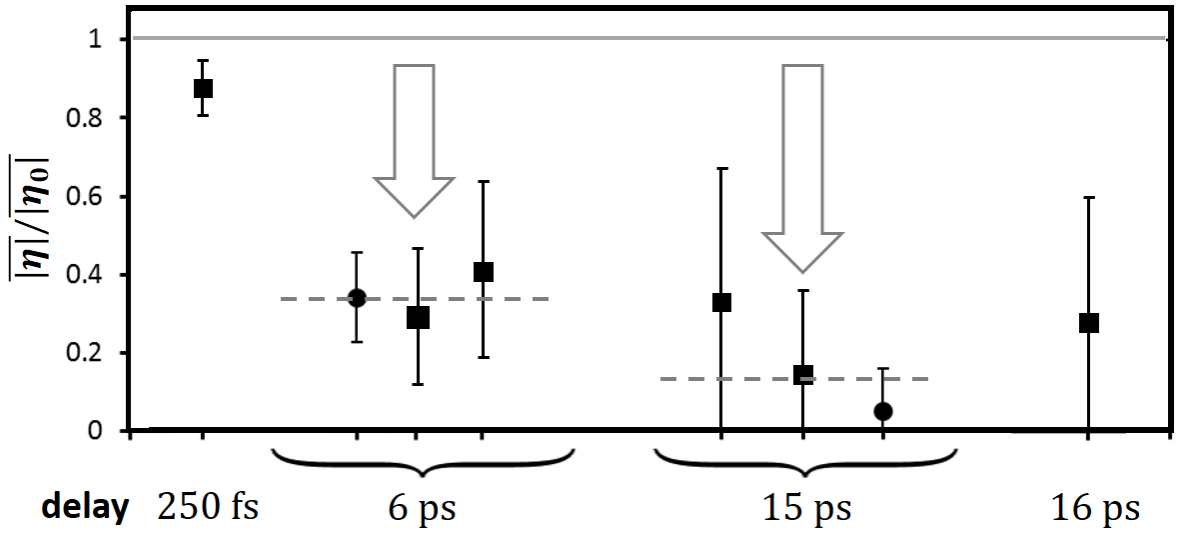


$$\eta_i = \frac{I(\vec{q_i}) - I(-\vec{q_i})}{I(\vec{q_i}) + I(-\vec{q_i})} \quad \rightarrow \quad \overline{\eta} \equiv \frac{1}{N} \sum_i^N |\eta_i|$$

Results: Asymmetry Analysis

- Asymmetry decreases in the pumped shots





asymmetry in pumped shot normalized by preshot

Methods: Asymmetry modeling

analytic target model

$$z_1(\boldsymbol{\sigma_1}), z_2(\boldsymbol{\sigma_2}), f''_{Si}, f''_{Cu}$$

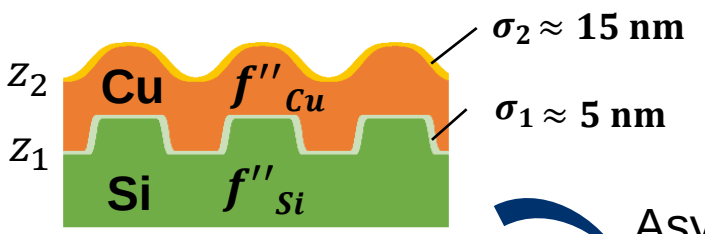
geometric parameters — material form factors

$$\eta_i = \frac{I(\vec{q_i}) - I(-\vec{q_i})}{I(\vec{q_i}) + I(-\vec{q_i})}$$

asymmetry model

$$\bar{\eta}(\boldsymbol{\sigma_1}, \boldsymbol{\sigma_2}, f''_{Si}, f''_{Cu})$$

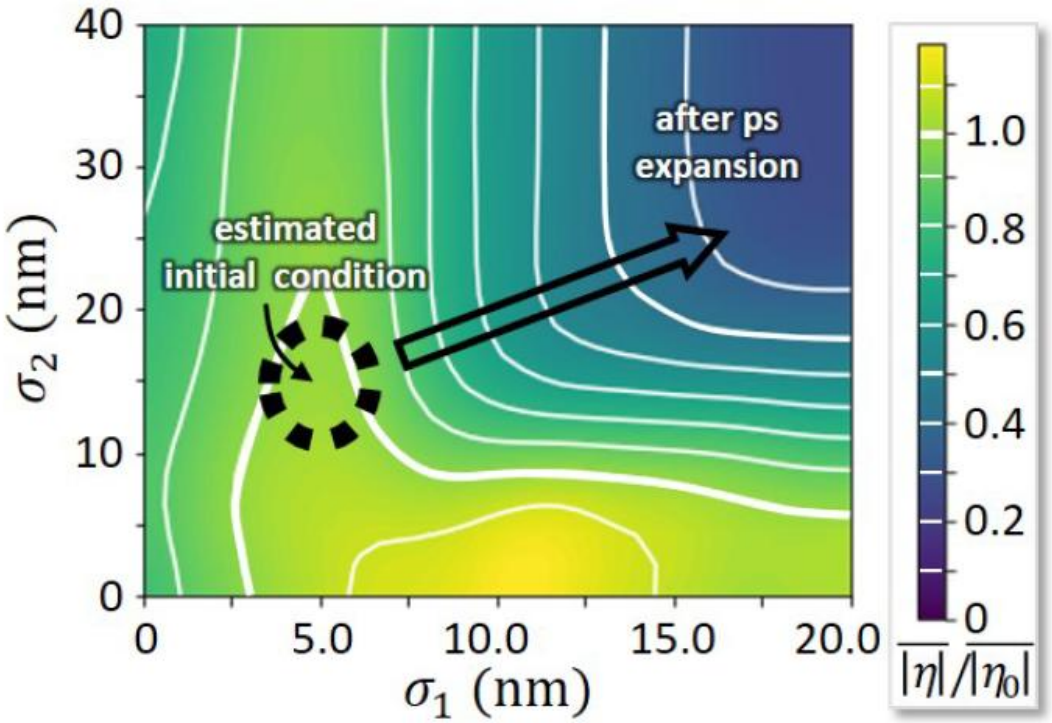
→ grating expansion parameters



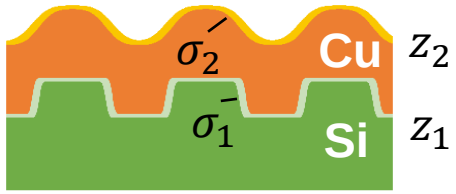
Asymmetry model

$$\bar{\eta}(\boldsymbol{\sigma_1}, \boldsymbol{\sigma_2}, f''_{Si}, f''_{Cu})/\bar{\eta}_0$$

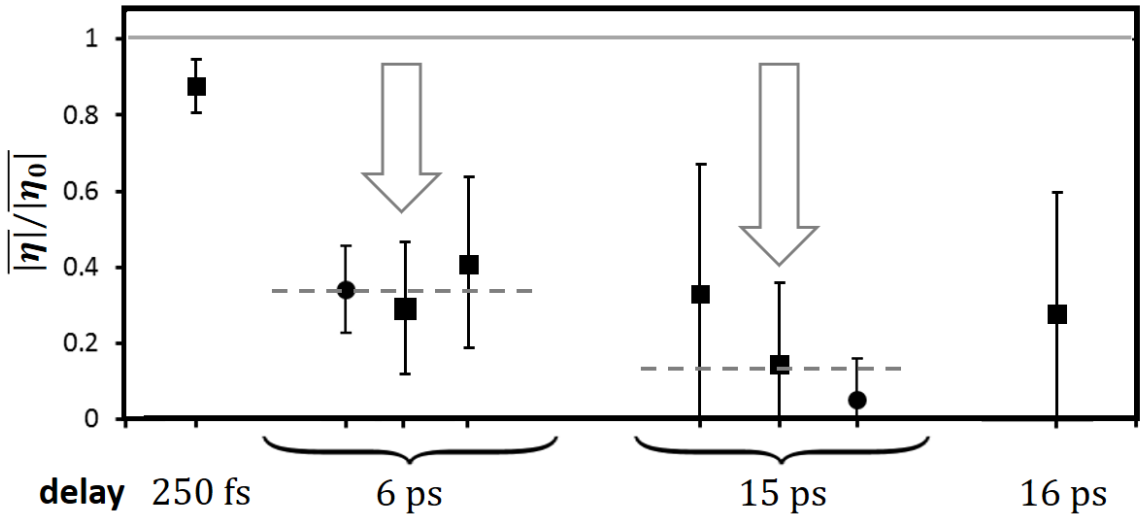
Modeled asymmetry



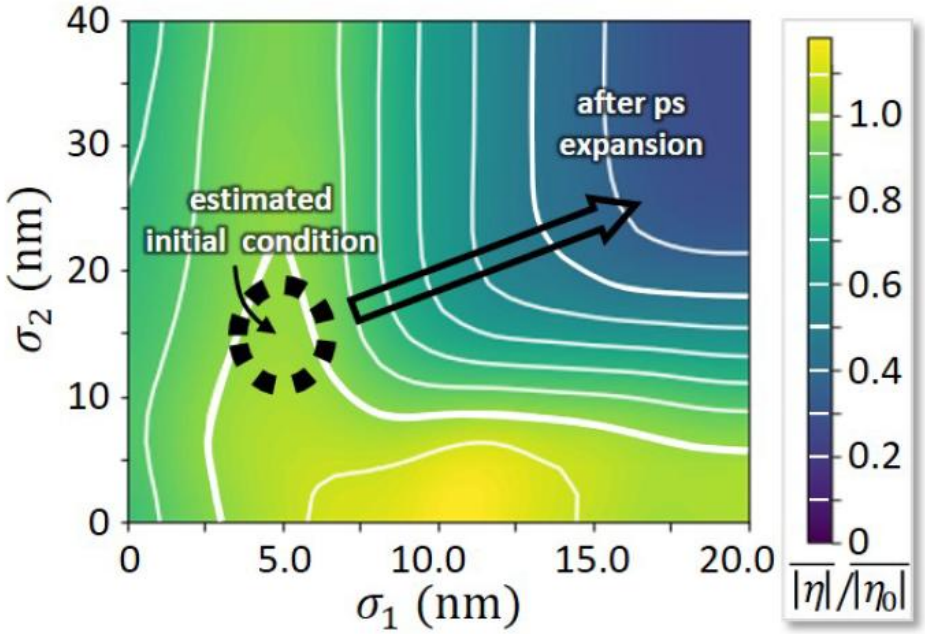
Results: Buried layer expansion



Experimental asymmetry measurement



Modeled asymmetry



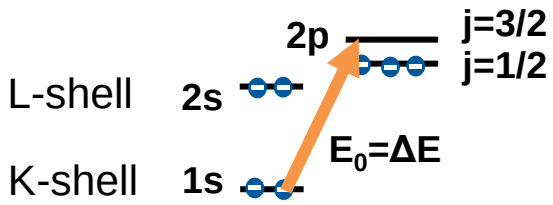
- Asymmetry decreases following laser pump

→ Asymmetry model reveals buried layer expansion

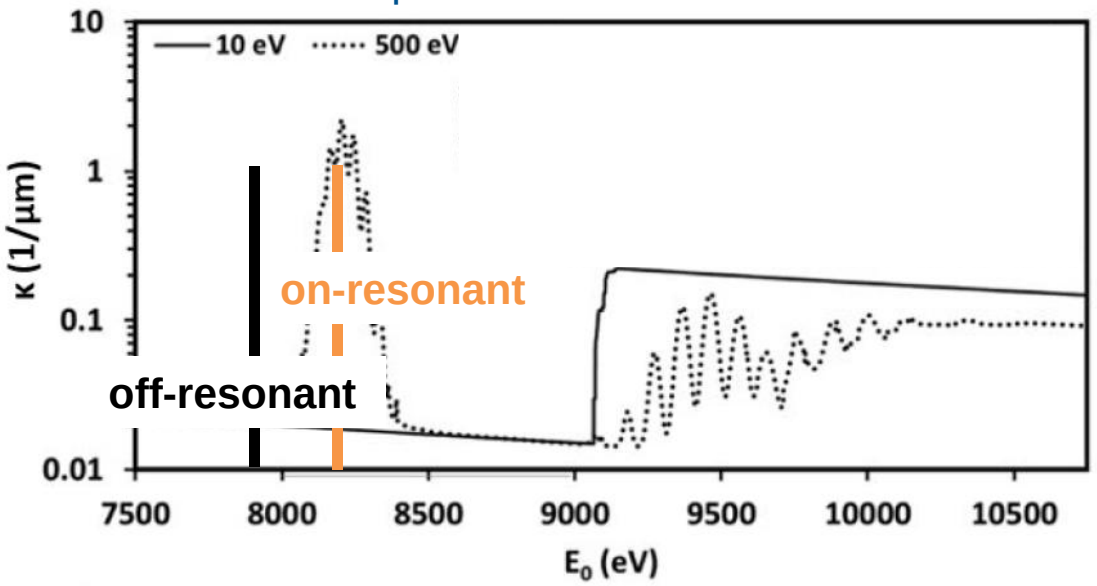
Experiment II: Can we also investigate the effect of absorption changes on the asymmetry?

Asymmetry model

$\bar{\eta}(\sigma_1, \sigma_2, f''_{Si}, \boldsymbol{f''_{Cu}})$

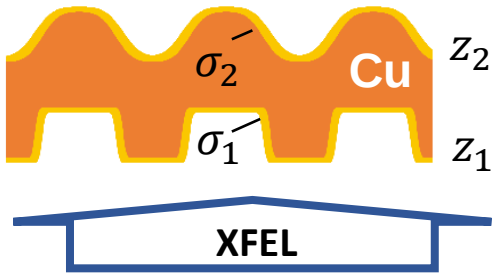


Absorption cross section of Cu



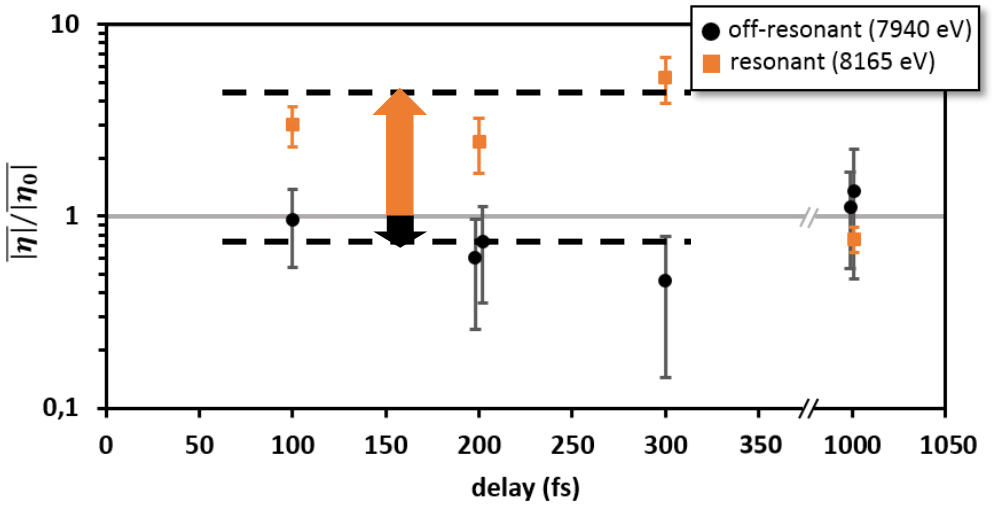
First resonant SAXS experiment at MEC

- Tune the XFEL to a resonant K_α transition energy in Cu
  - 8165 eV → resonant transition in Cu ions with 7 bound electrons in the inner shells (N-like)
- Increase of pump laser intensity to 10^{19} W/cm^2
- Direct irradiation of the Cu layer



Results II: Ultrafast ionization measured with Resonant SAXS

Experimental asymmetry measurement



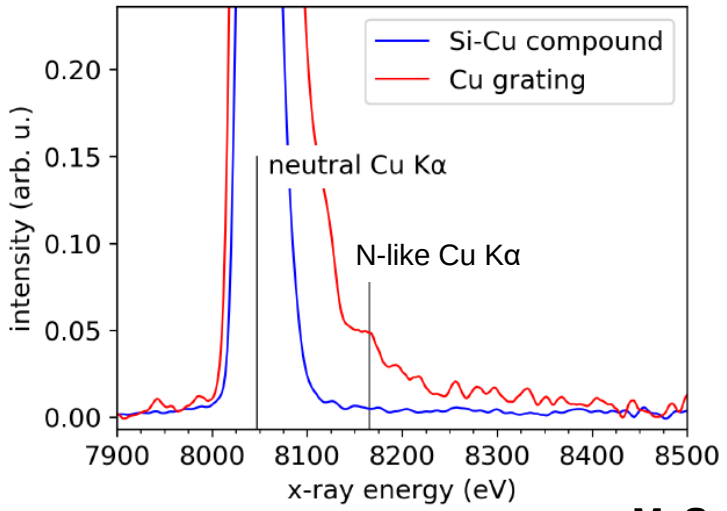
Observation:

- **off-resonant**: slight decrease in asymmetry
- **on-resonant:** increase in asymmetry during first 300 fs

Interpretation:

- Asymmetry reflects absorption changes due to **N-like ionization** in the hot Cu plasma

K_α- emission spectroscopy confirms presence of N-like ionization

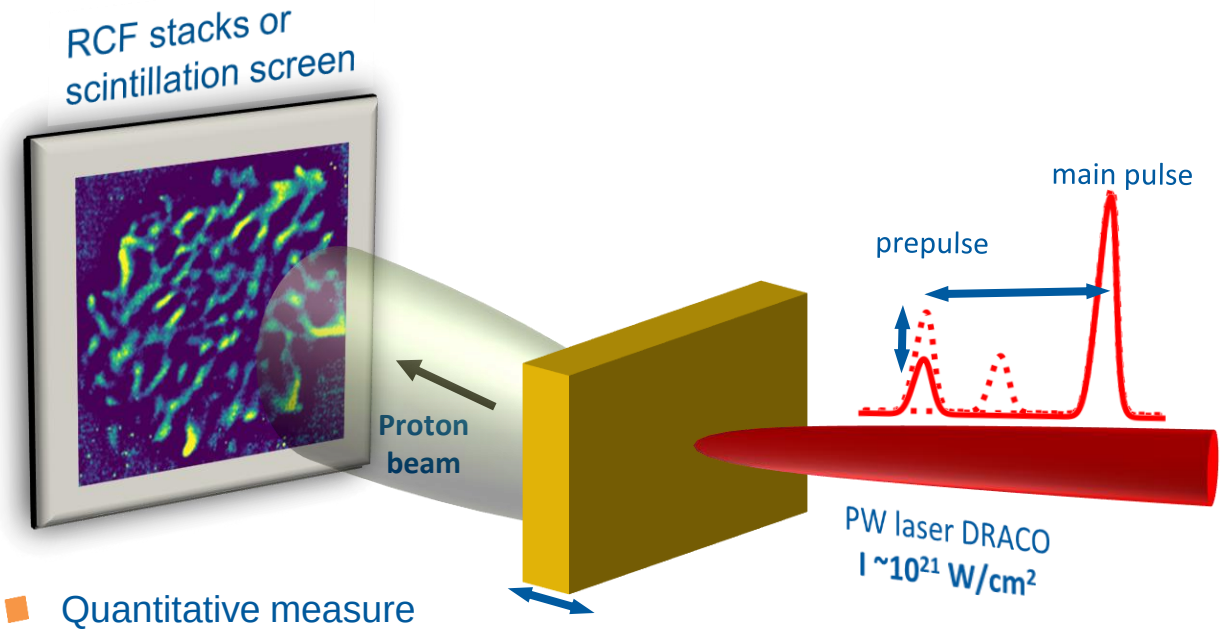


M. Smid

→ Resonant SAXS gives access to the transient ionization physics during Laser-Matter Interaction!

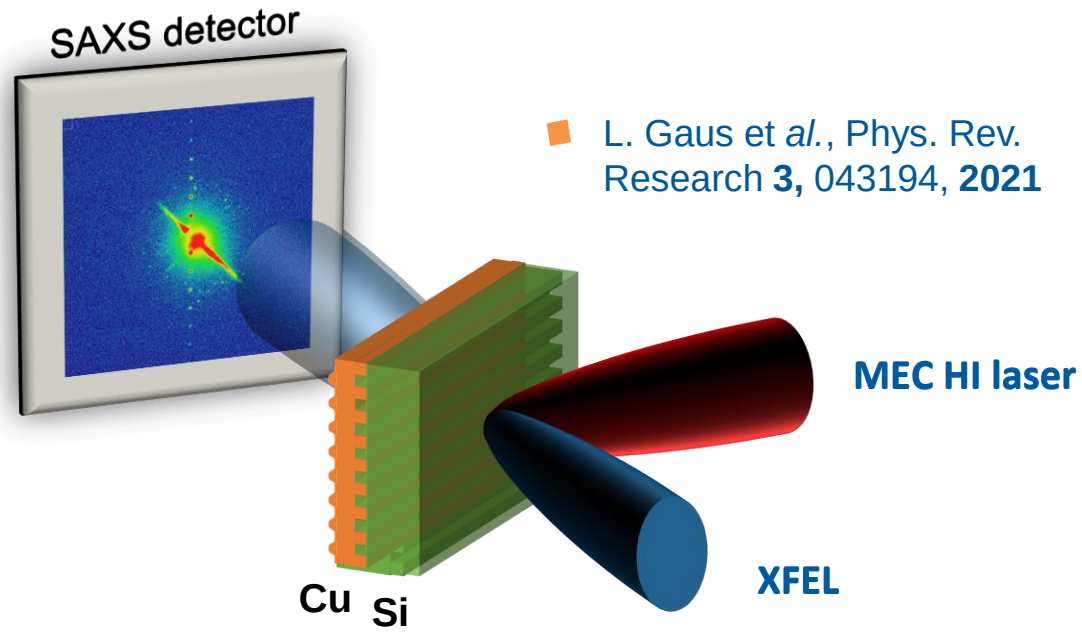
Summary of my PhD project

1) UHI laser-only testbed experiments



- Quantitative measure of modulation strength
- Variation of target **material, thickness, composition**
- Variation of laser **contrast** & laser pulse **energy**

2) Demonstration Experiments of SAXS at MEC

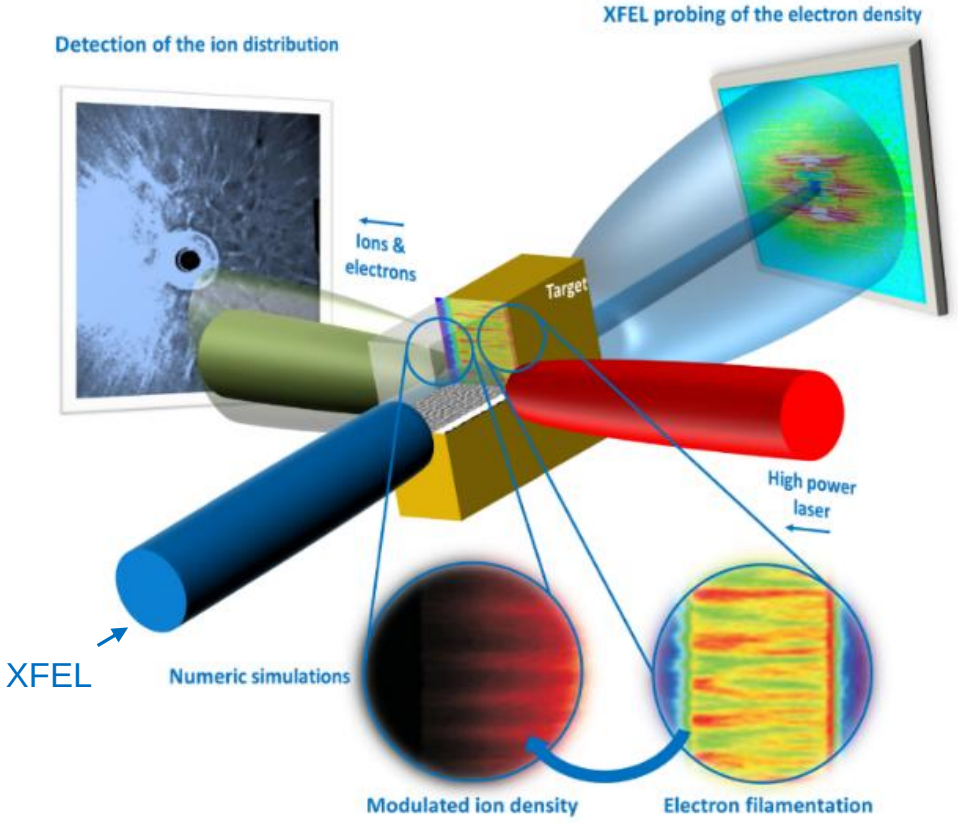


- L. Gaus et *al.*, Phys. Rev. Research **3,** 043194, **2021**
- Buried layer expansion
  - access to nm scale
- Ionization via Resonant SAXS
  - access to fs time scale

Outlook

- Benchmark ionization models in PIC simulation codes far from equilibrium
  - Multicolor X-ray probing

- Study laser-induced plasma structures with (resonant) SAXS



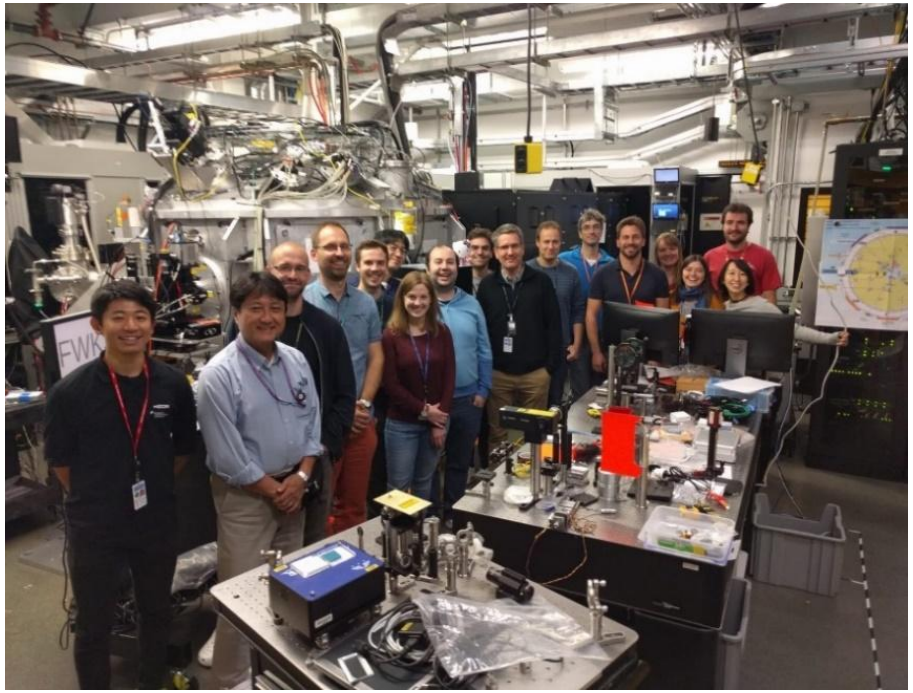
HZDR

HiBEF

Thanks for your attention!

And many thanks to…

- Prof. U. Schramm, Prof. T. Cowan
- Dr. Karl Zeil, Dr. Thomas Kluge
- The LT94 team
- All HZDR colleagues
- Friends and family



LT94 Experiment Team

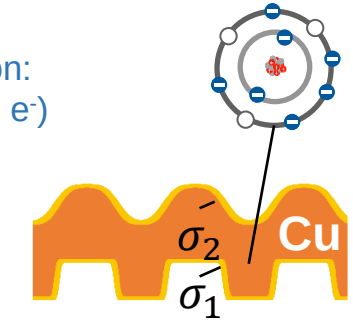
Backup

Summary

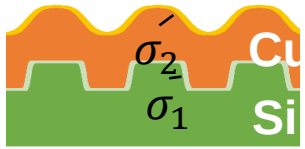
asymmetry model

$\eta(\sigma_1, \sigma_2, f''_{Si}, f''_{Cu})$

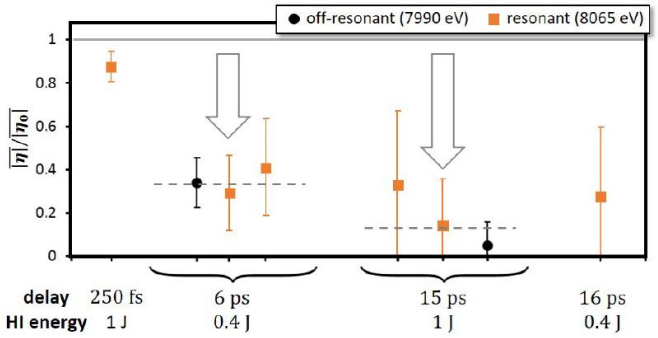
O-like ionization:
Cu^{22+} (7 bound e^-)

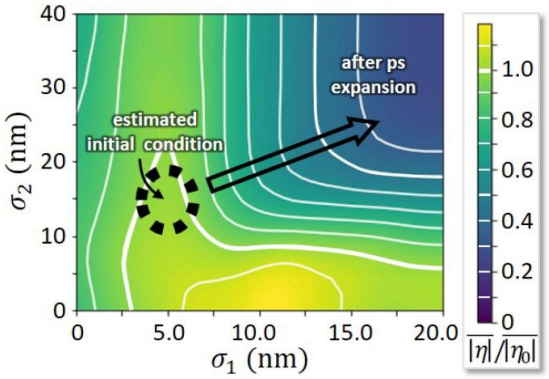


Buried layer Expansion



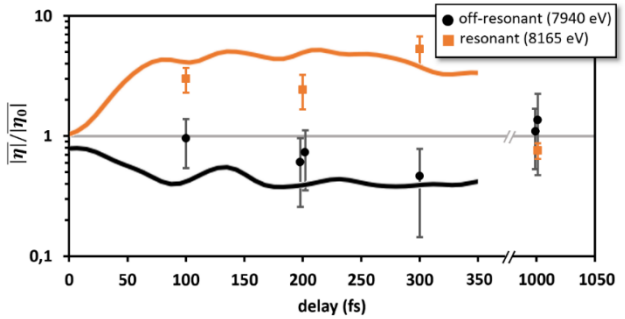
- Asymmetry decrease indicates buried layer expansion

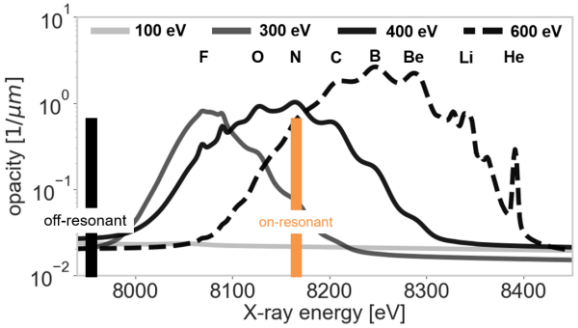




Ultrafast opacity changes measured with resonant SAXS

- Asymmetry increase due to O-like ionisation in copper plasma

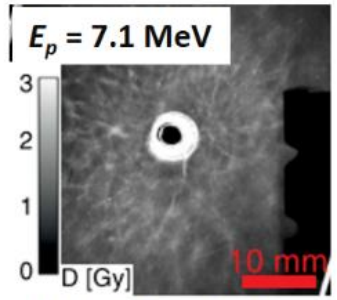




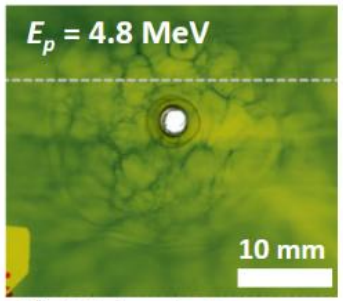
A plethora of mechanisms in the literature

Weibel-like instability

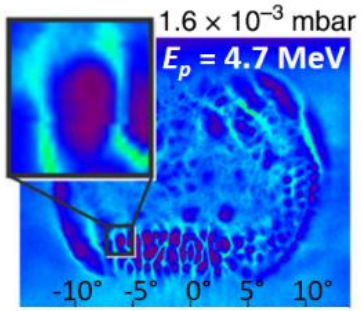
Rayleigh-Taylor instability or Weibel-like instability



a) Metzkes et al.



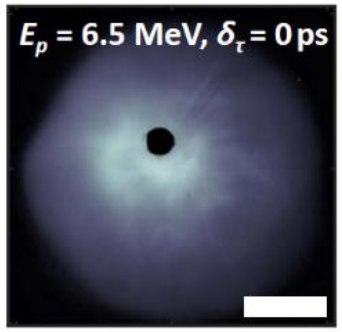
b) Göde et al.

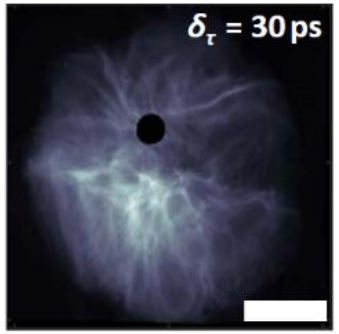


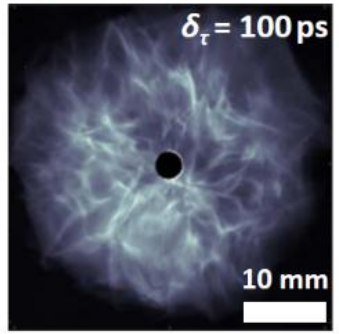
c) Obst-Hübl et al.

All-optical imprinting

Weibel-like instability

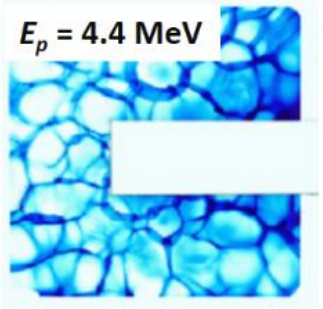




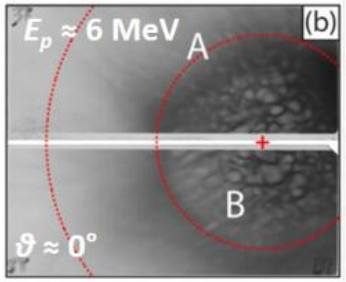


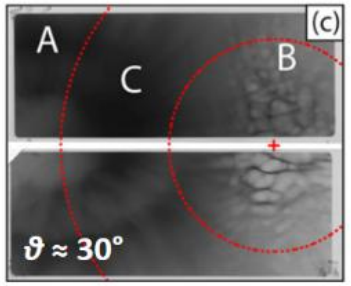
d) Scott et al.

RPA & RTI



e) Palmer et al.

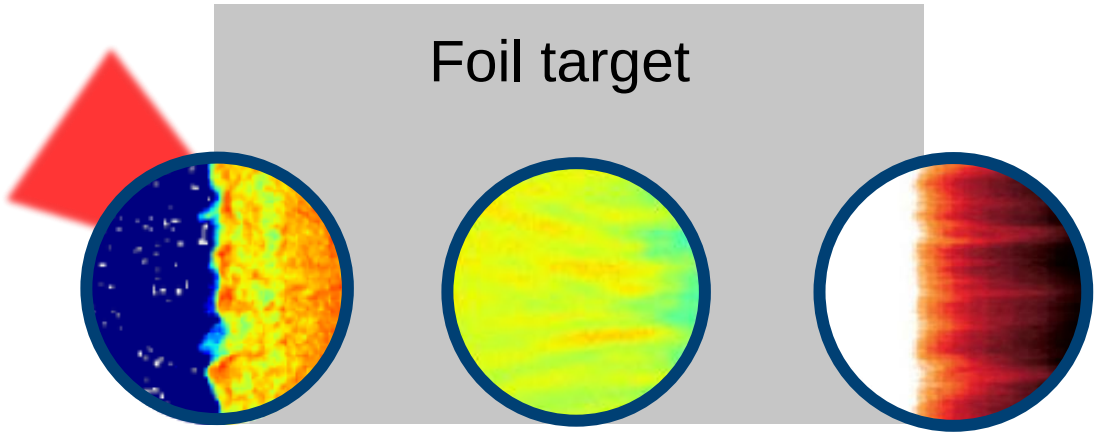




f) Powell et al.

RPA & RTI

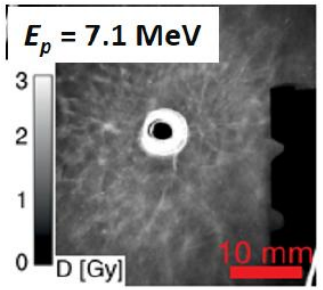
A plethora of mechanisms in the literature

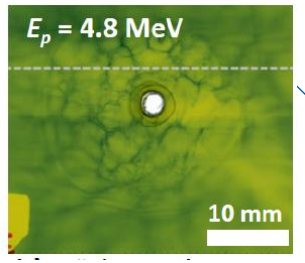


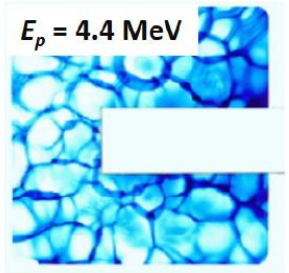
front side:
RT-like instability,
RPA +RTI

Plasma bulk:
Weibel-like instability

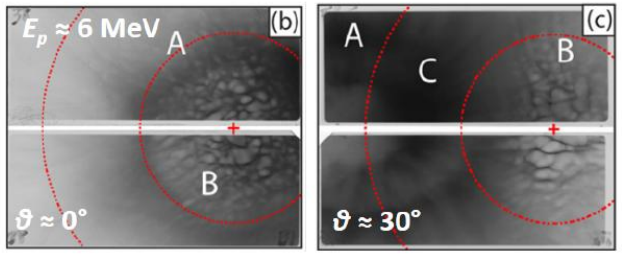
Rear side:
Weibel-like instability



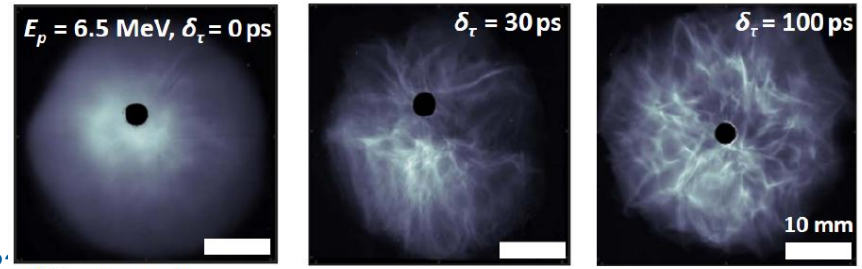




e) Palmer et al.

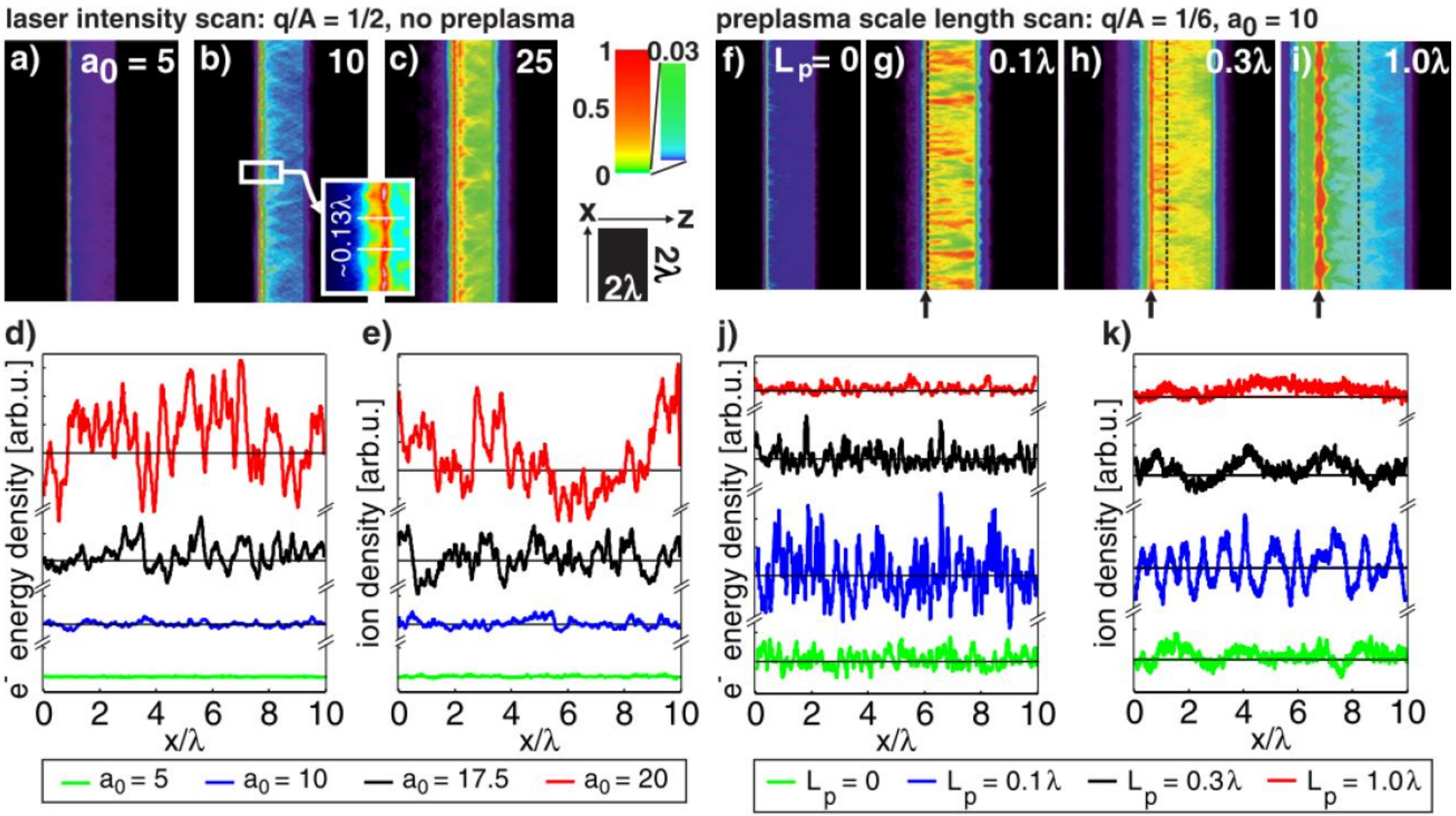


f) Powell et al.

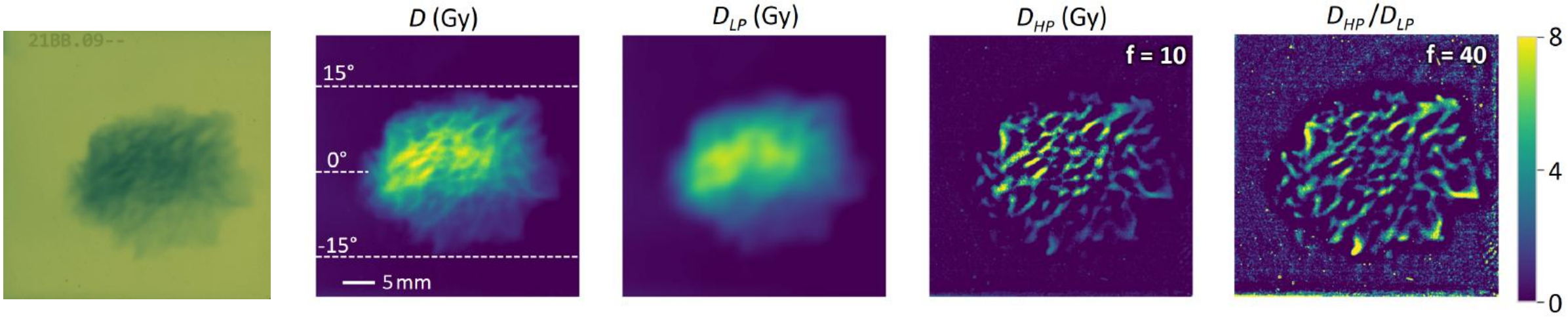


d) Scott et al.

NJP, Metzkes-Ng et al. - PIC simulations



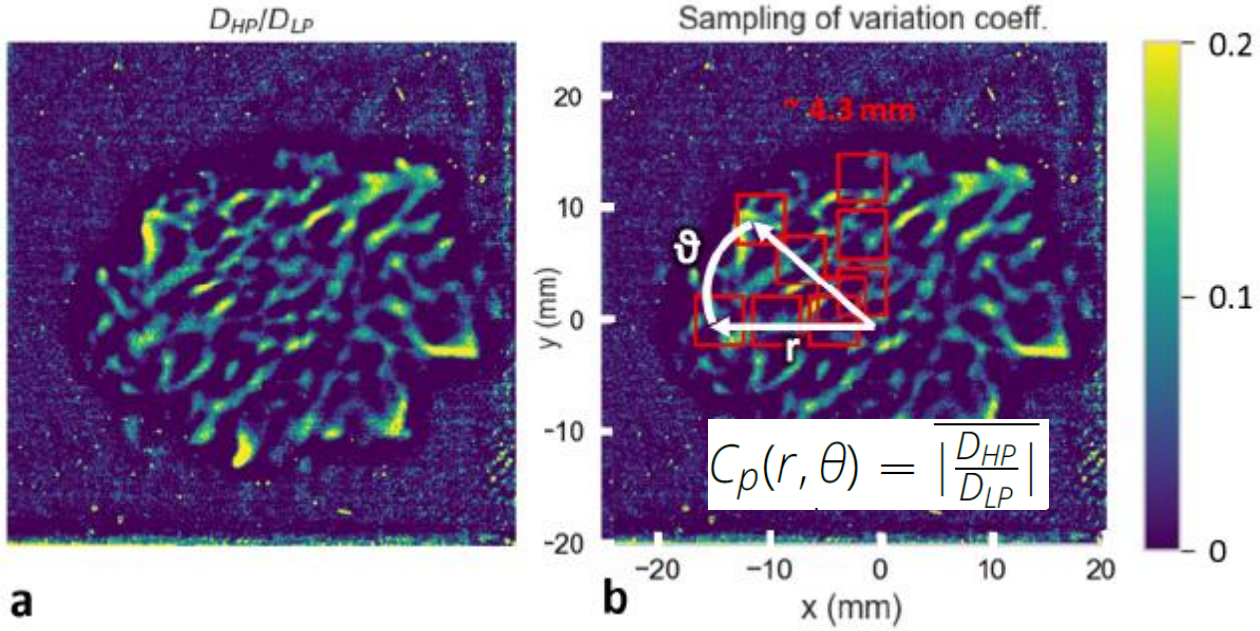
Data Processing to highlight beam modulations

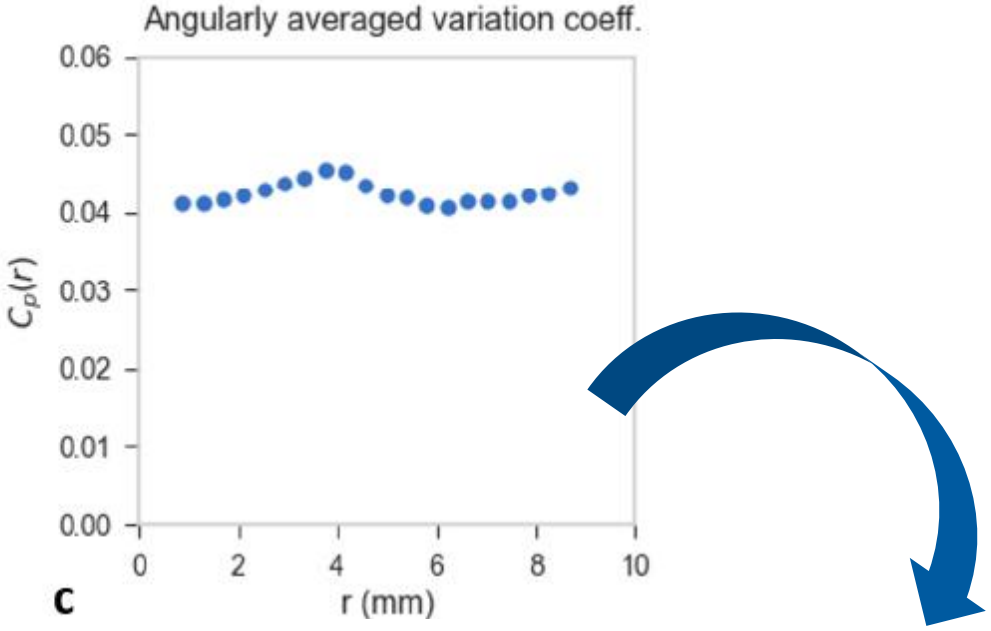


raw RCF data, FV 120 nm

Image processing to carve out high frequency modulations

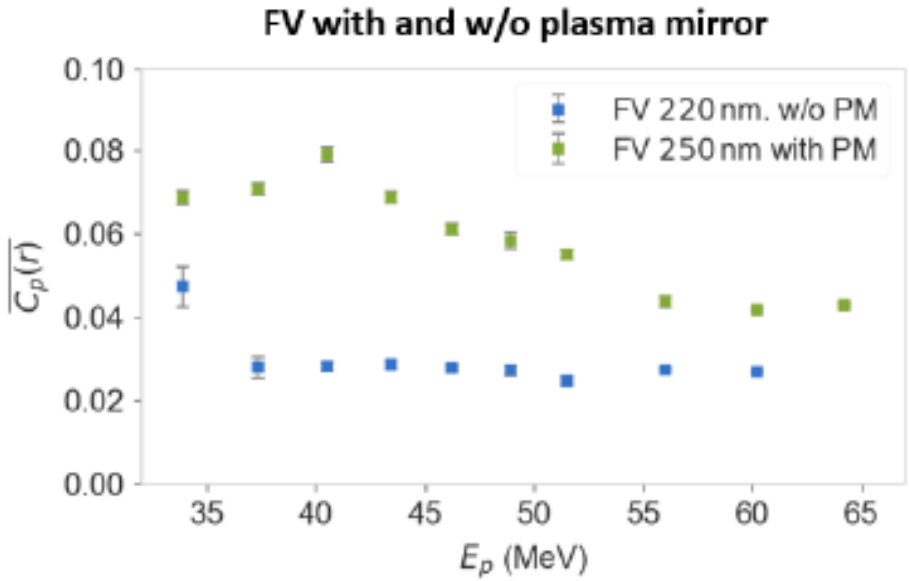
Quantitative measure of the modulation strength





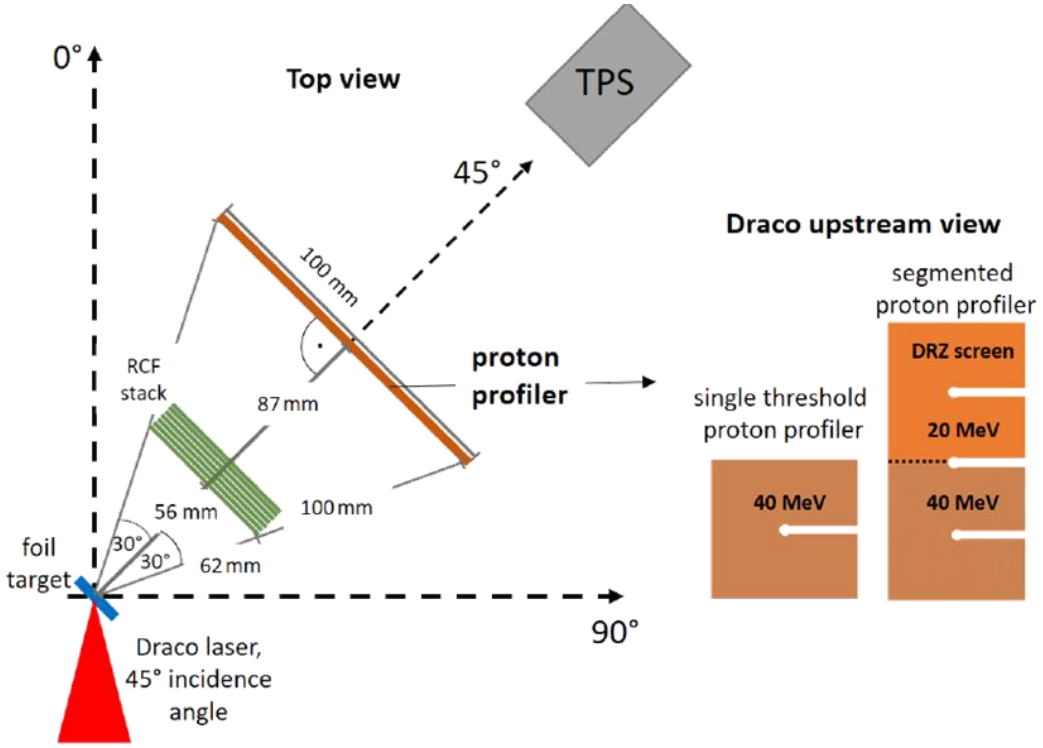
High pass filtered dose profile normalized by low pass filtered profile

- Data sampling a la MacLellan
- Variation coefficient C_p in ROIs

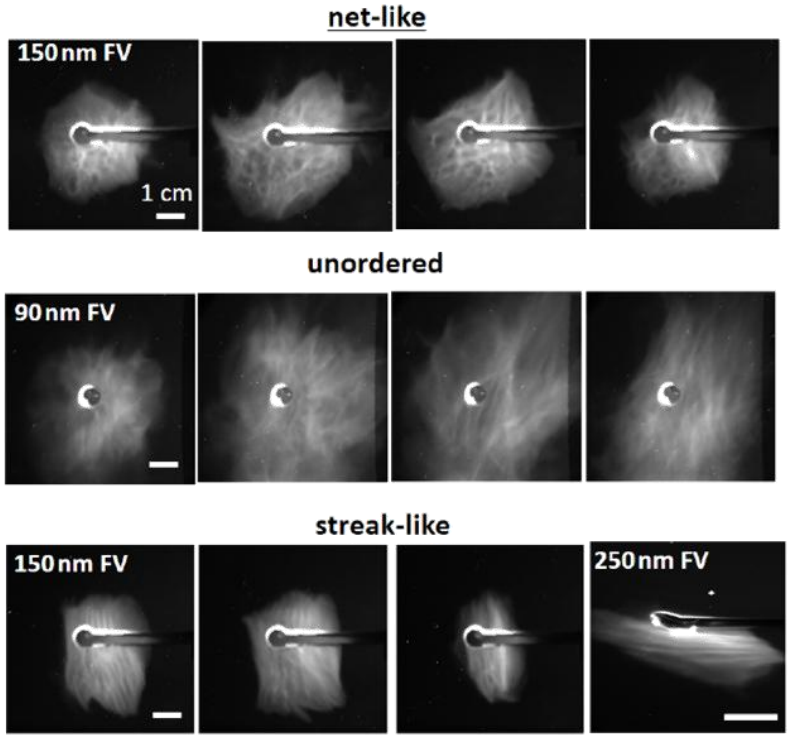


Experimental study of transverse proton beam modulations

Experimental Setup



Examples of different beam profile types



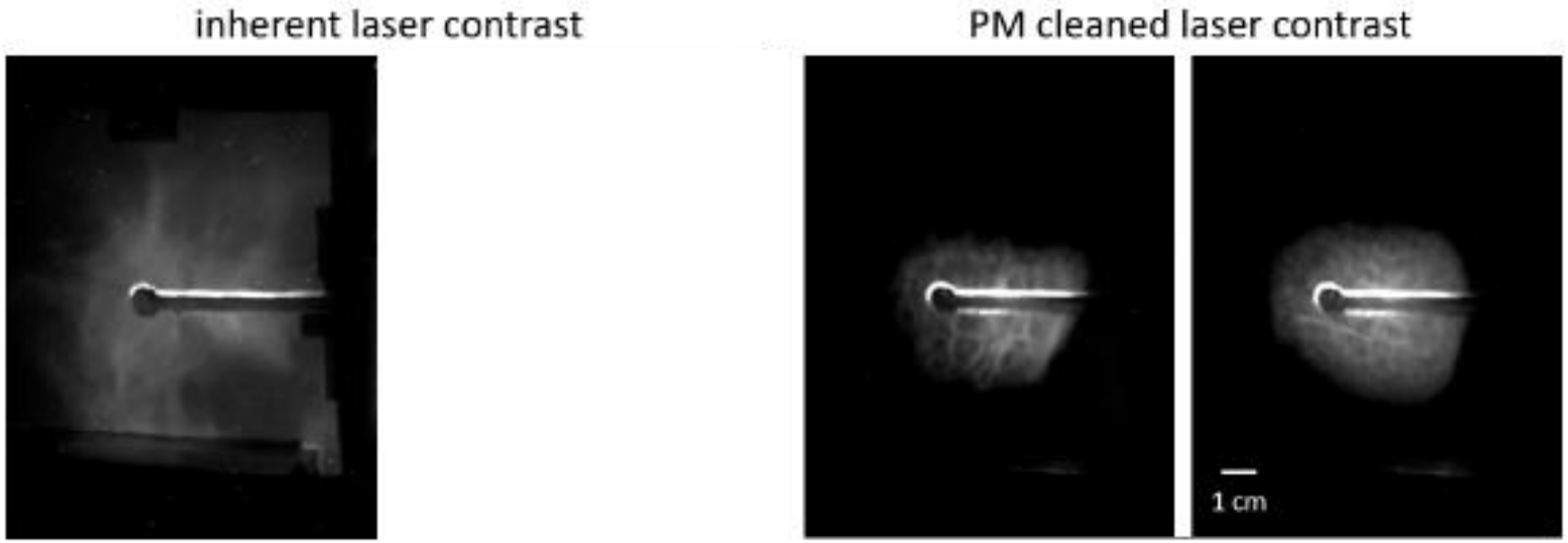
Empirical observations:

- Net-like p^+ beam modulations occurred with few 100 nm FV targets at **plasma mirror cleaned laser contrast**
- No transmitted light

Further studies at optimized contrast:

- Target thickness scan
- Laser energy scan
- Al-coated FV targets at front or rear side

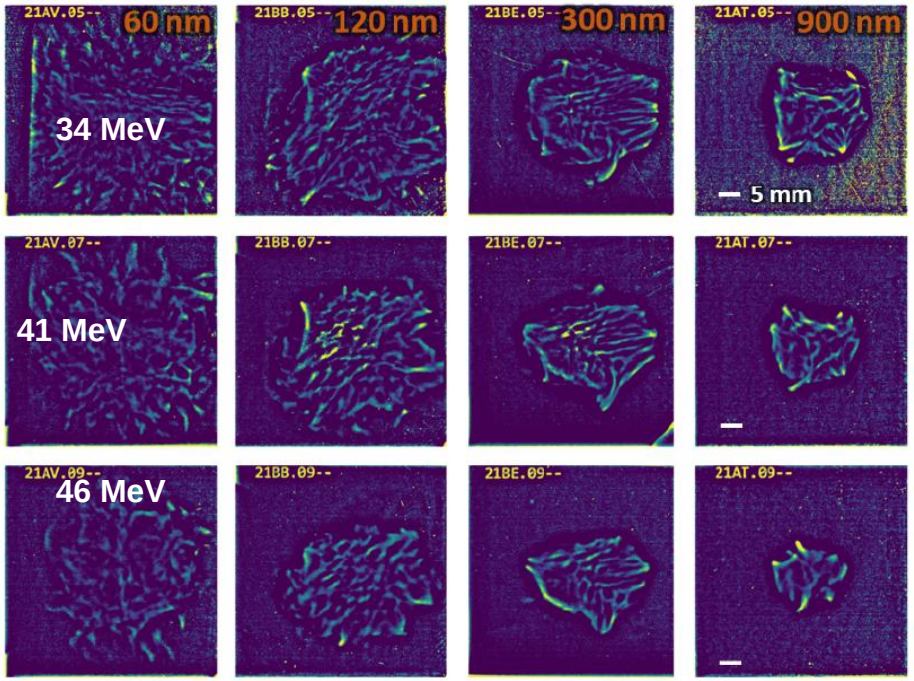
Plasma Mirror is virtually a switch on for net-like beam generation



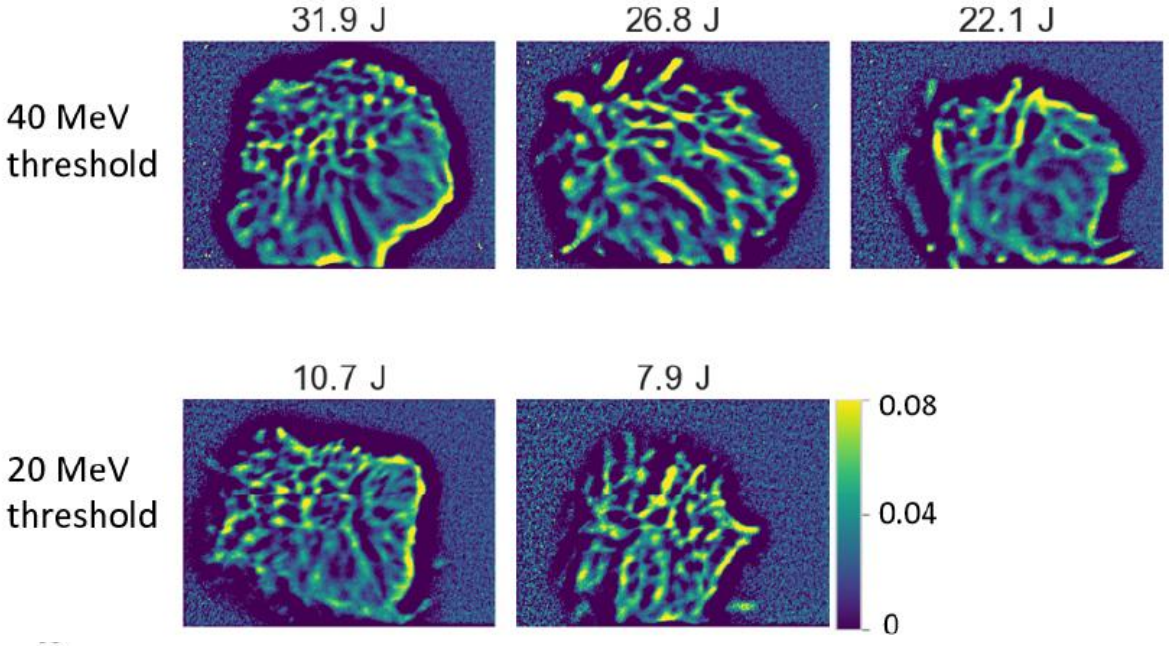
- data from **one** shot day
- FV foils, 150 nm thickness

Experiments at plasma-mirror cleaned laser contrast

FV thickness scan - RCF data (HP filtered)



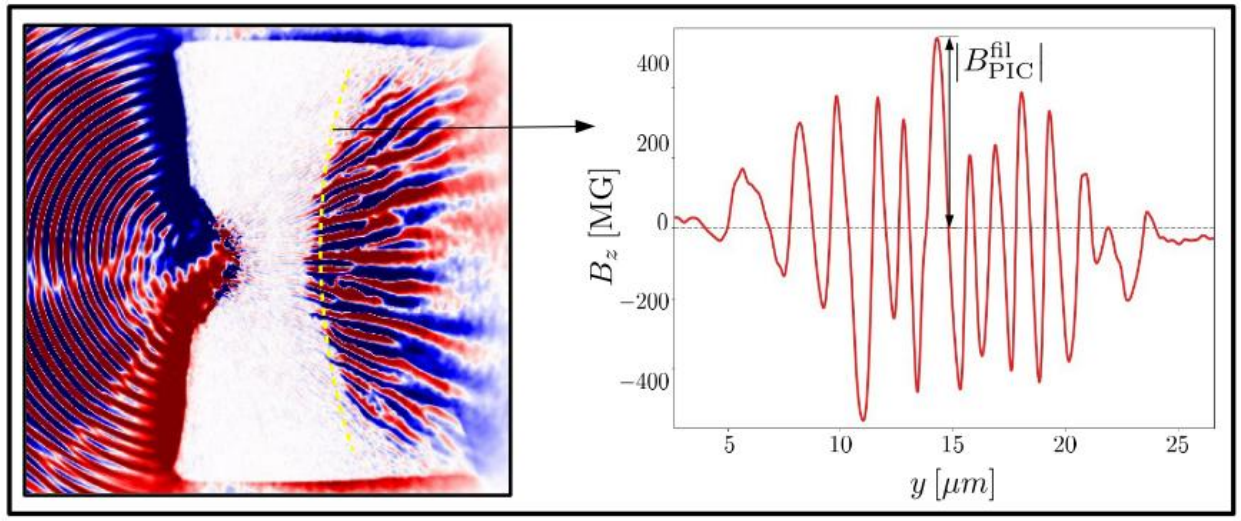
Laser energy scan - Lanex measurements (HP filtered)



- Modulated structures occur in …
  - a wide thickness range
  - within the whole beam spot area
  - the whole spectrum
  - target normal direction

- Modulations in the experimental range of laser energy
- E_p follows square root scaling with E_L

Proton deflection in transverse magnetic fields are a possible explanation

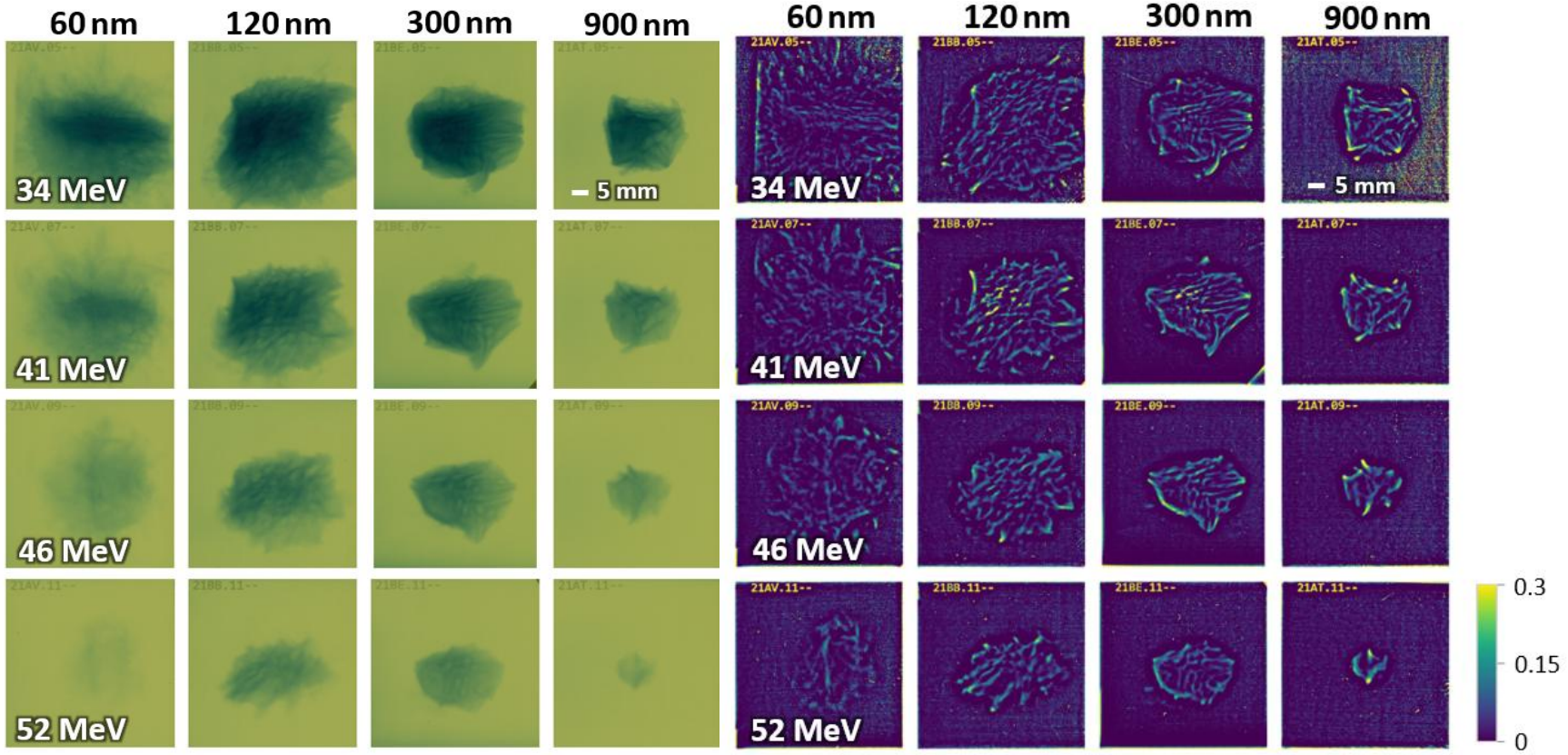


Filamented magnetic field structures, PIC simulations (J. Branco)

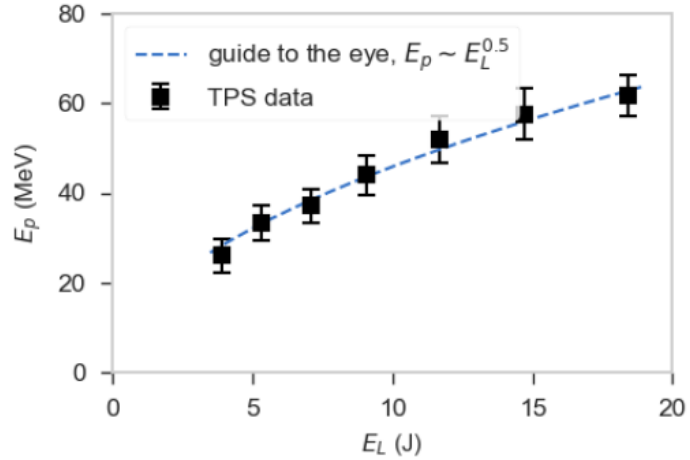
A possible interpretation:

- Rear sheath accelerated protons probe modulated magnetic field structures at the target rear side

Formvar - thickness scan



cut-off energies

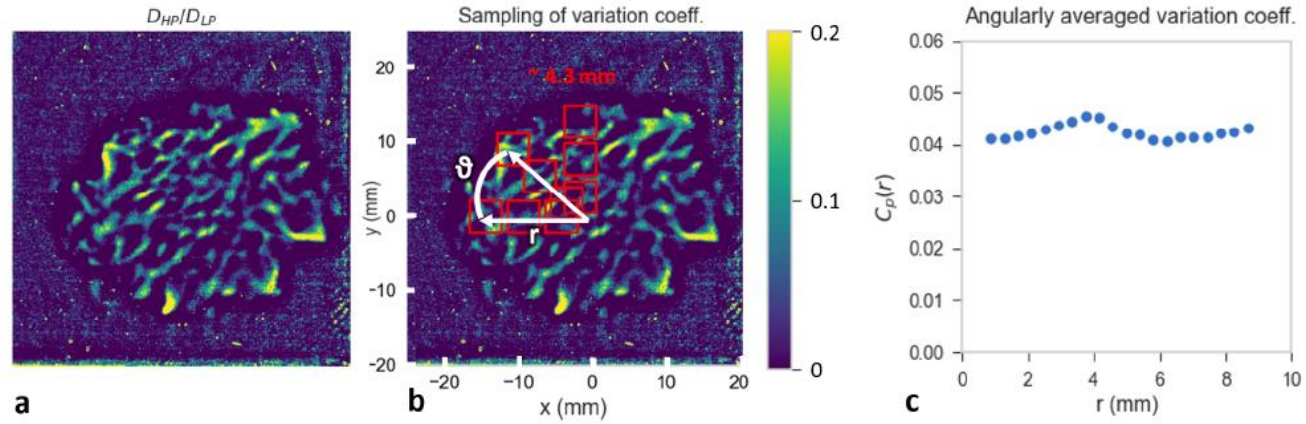


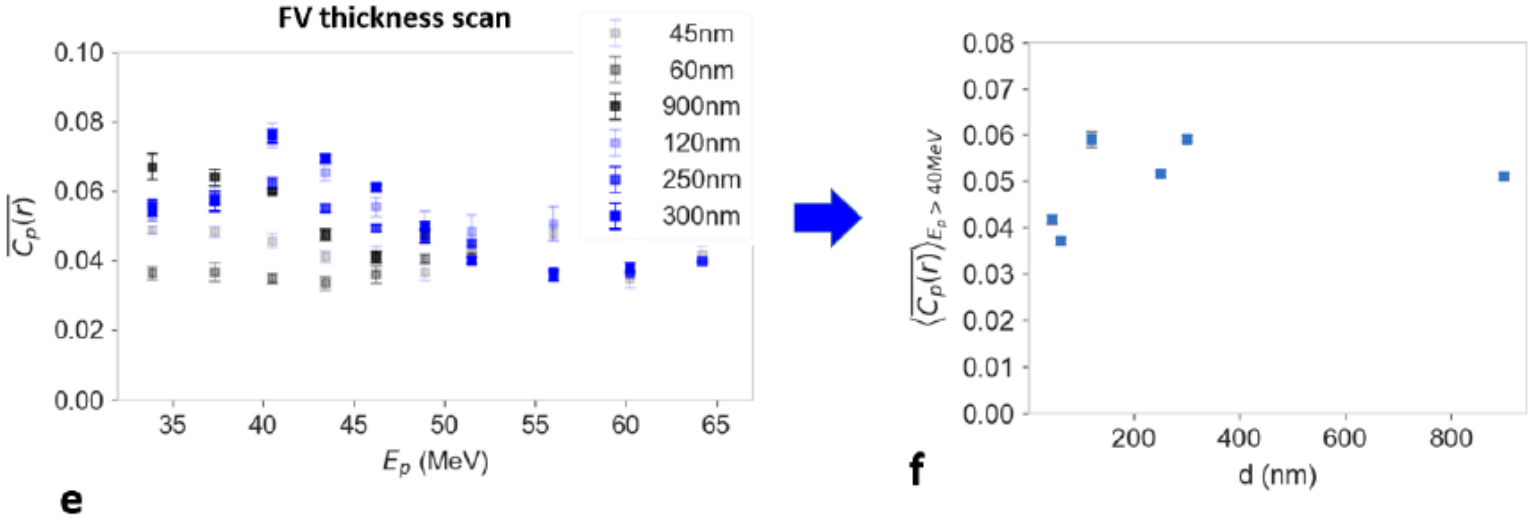
Modulations are…

- visible in the whole thickness range
- in the whole stack
- completely within the beam spot

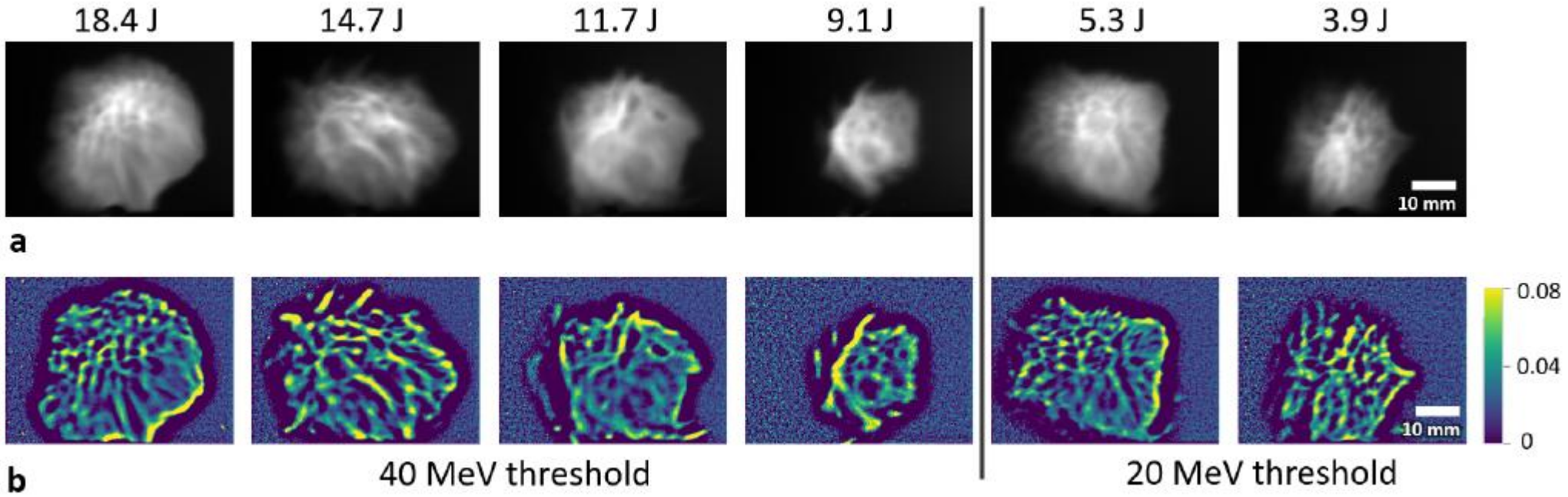
Formvar - thickness scan;

→ quantitative analysis of the modulation strength





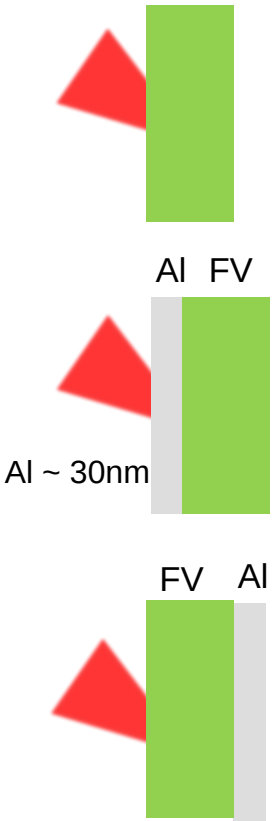
Experiments with Formvar targets; laser energy scan



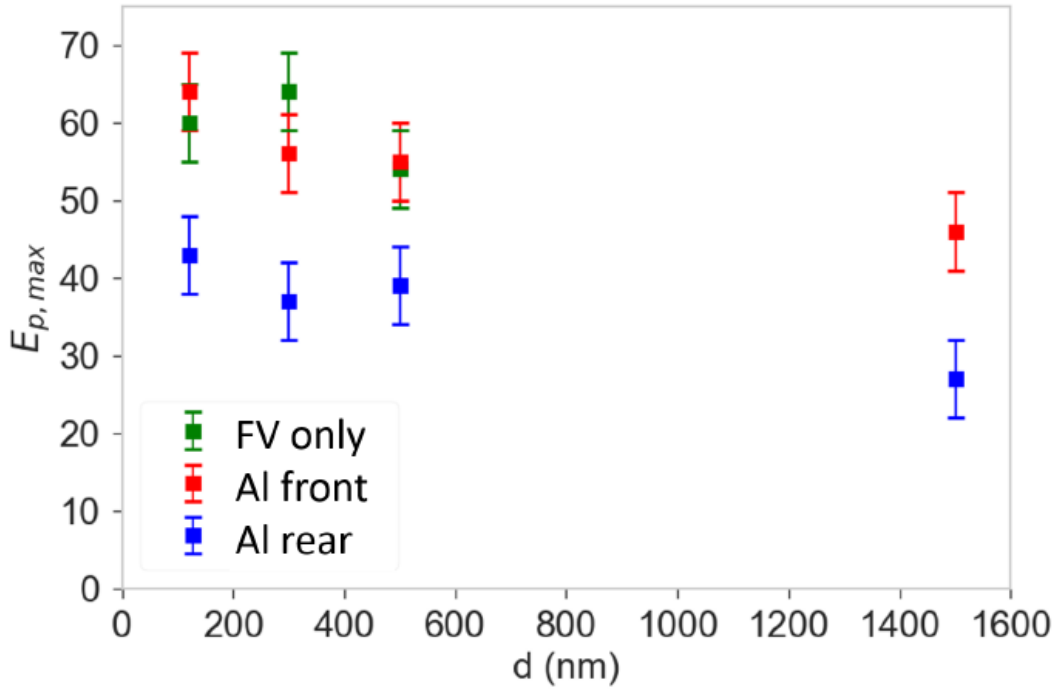
- Modulations are visible in the whole E-scan range

Experiments with Aluminium-coated Formvar targets

FV /plastic, variable thickness

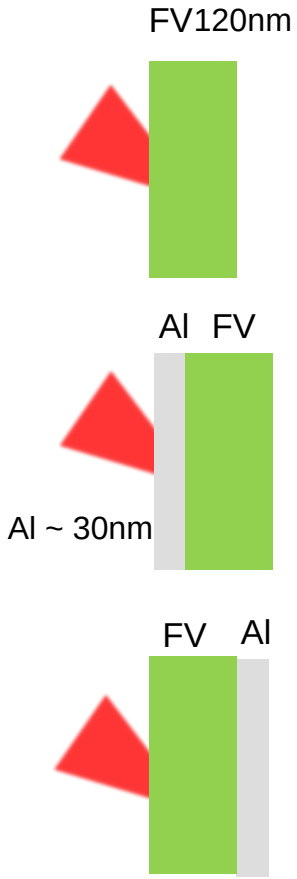


cut-off energies



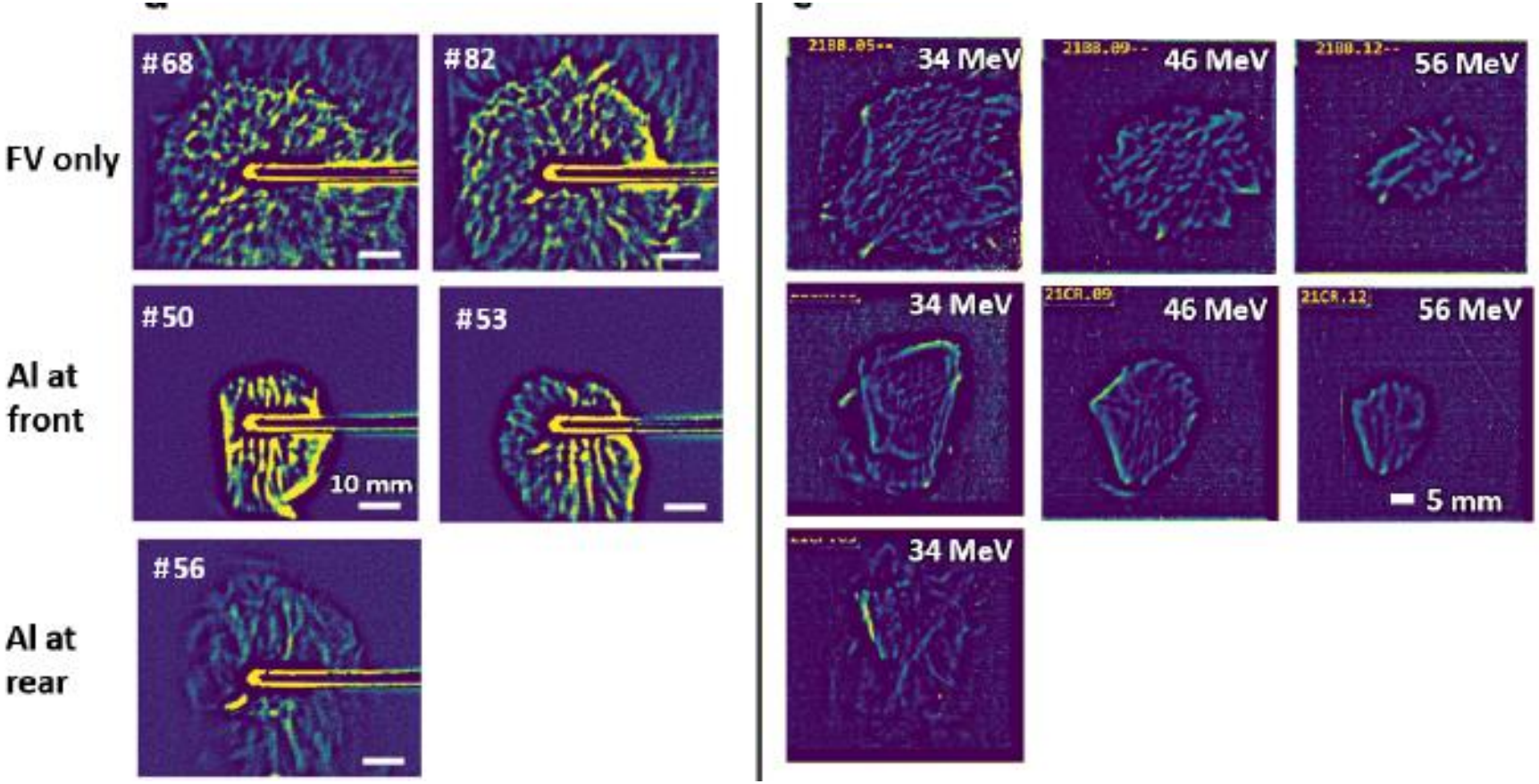
- Al at rear: acc. breaks down by 30%
- Al at front: similar acc. Performance as FV-only

Experiments with Al-coated Formvar targets



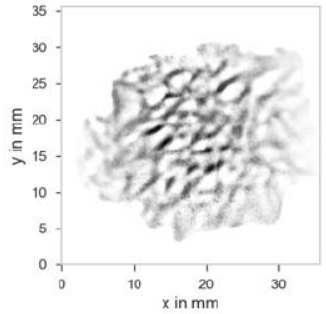
Lanex data

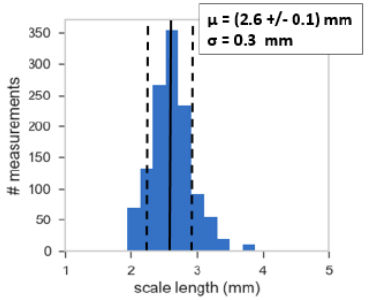
RCF data



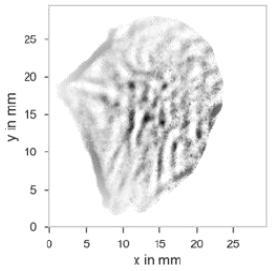
Al-coated Formvar targets → Modulation scale length analysis

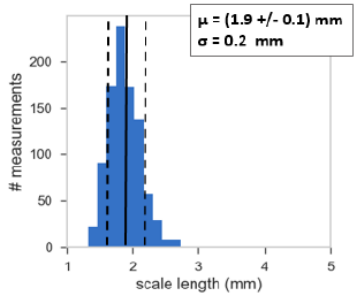
FV only
120 nm



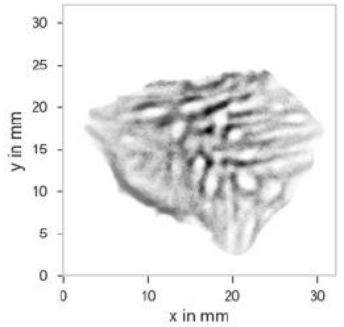


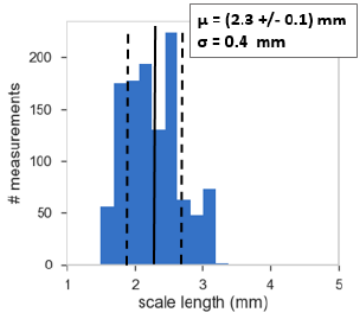
FV 120 nm
+ Al at front



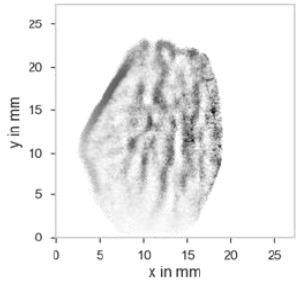


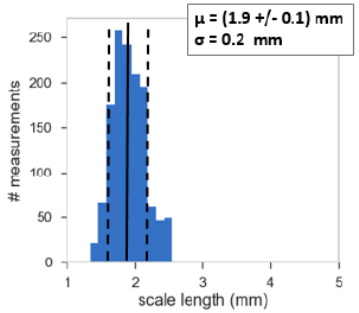
FV only
300 nm





FV 300 nm
+ Al at front

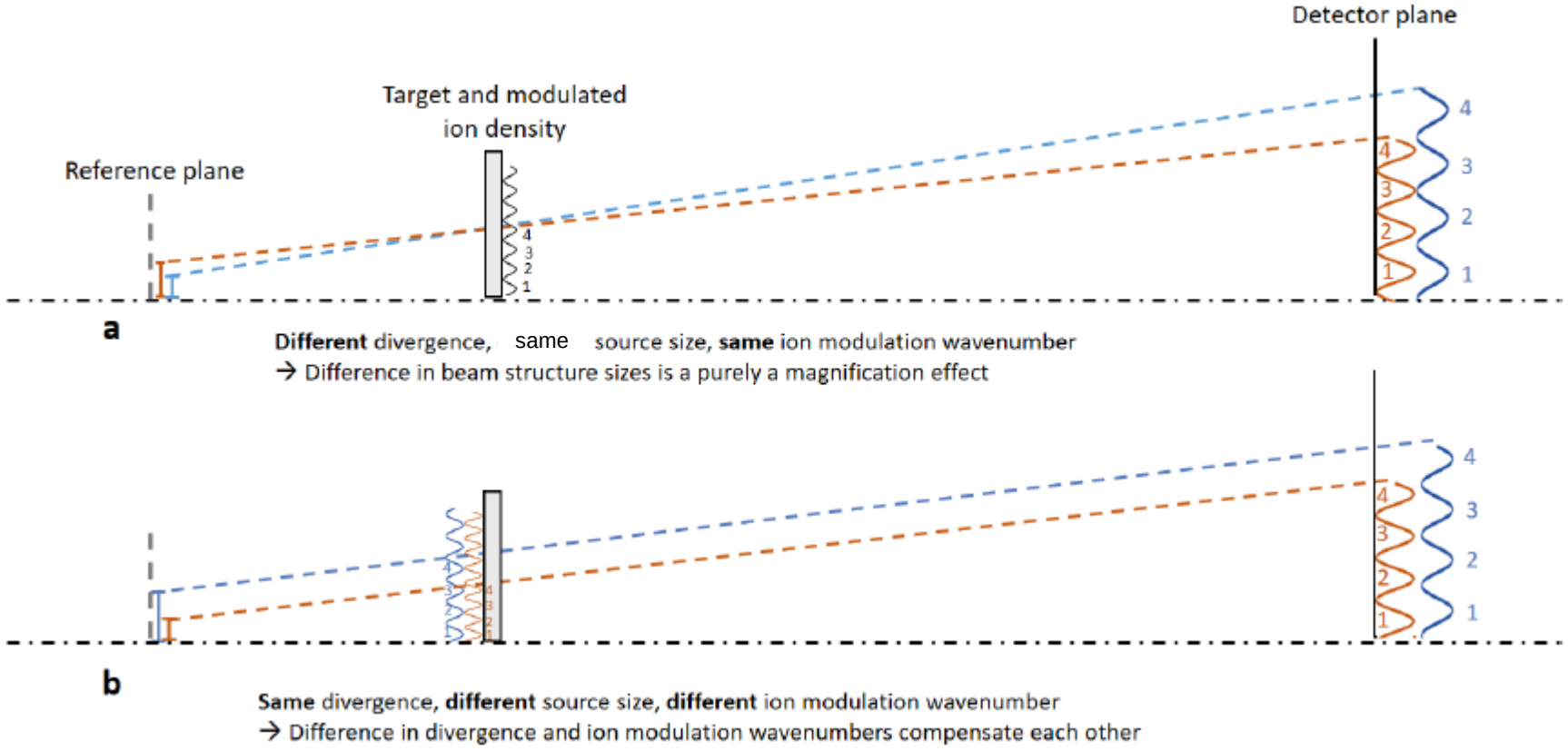




- FV-Al$_{front}$ generates smaller modulation scale lengths than FV-only
- But FV-Al$_{front}$ generates also smaller beam spot areas
  - These effect compensate each other, i.e. #modulations per beam spot size is equal in both configs.

Experiments with Al-coated Formvar targets

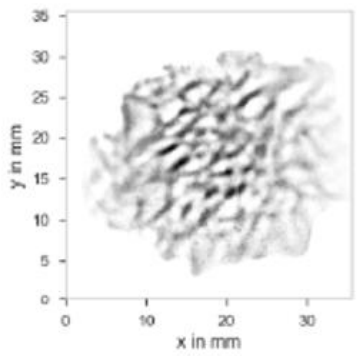
→ Different mechanism may explain same number of modulations in the beam sport

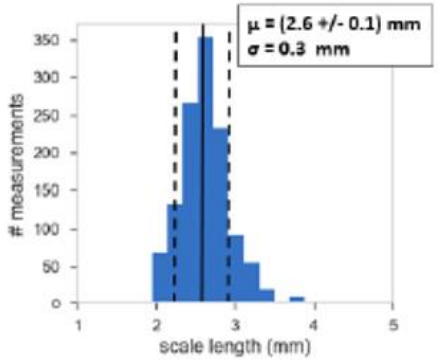


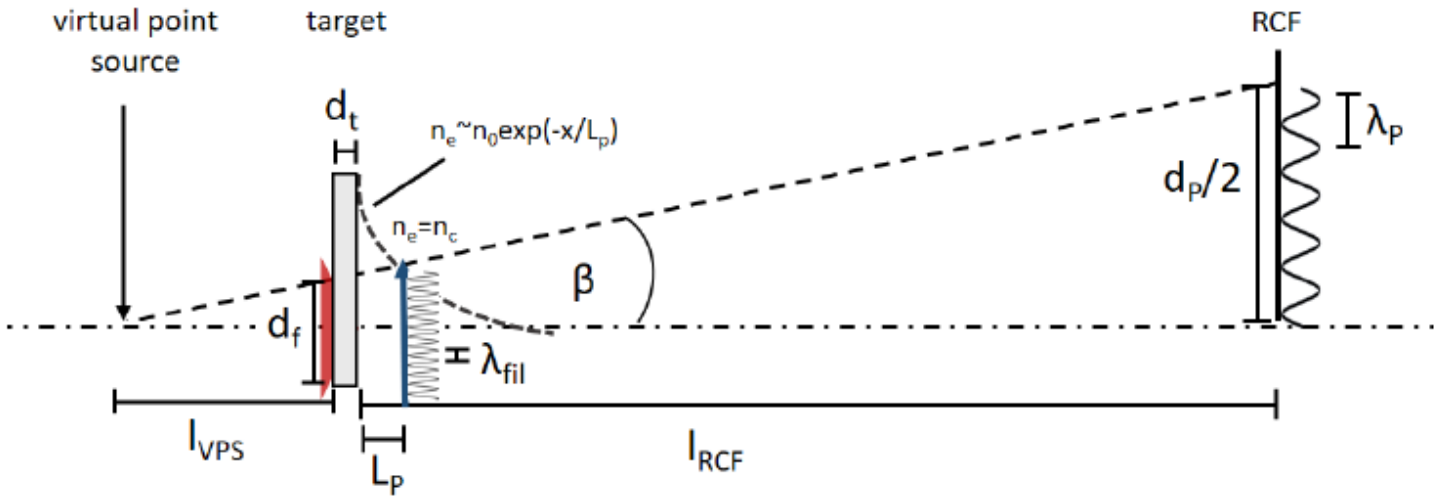
- We need to know the source size!
  - Follow-up experiments: use of prestructured rear surface a la Cowan et al.

Experiments with Formvar targets; laser energy scan

FV only
120 nm







$$\lambda_W = \frac{2\pi c\sqrt{\gamma_r}}{\omega_r}$$

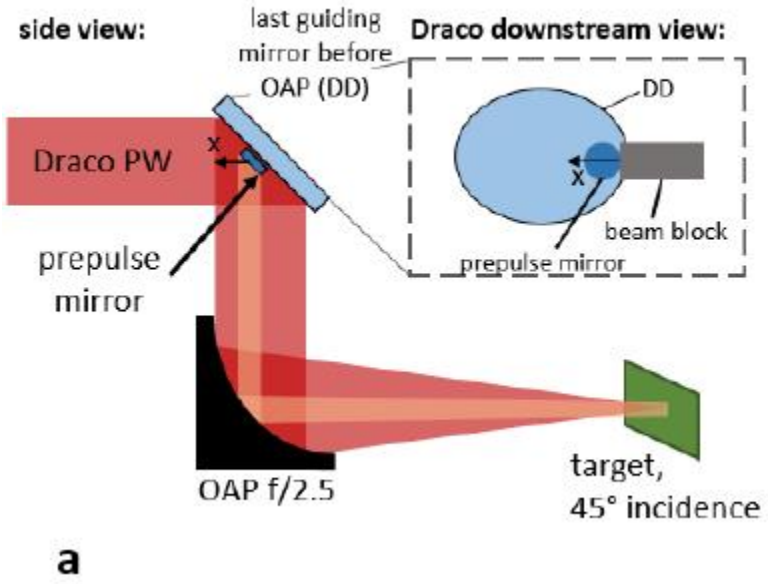
$\lambda_W = 4.8\,\mu m.$

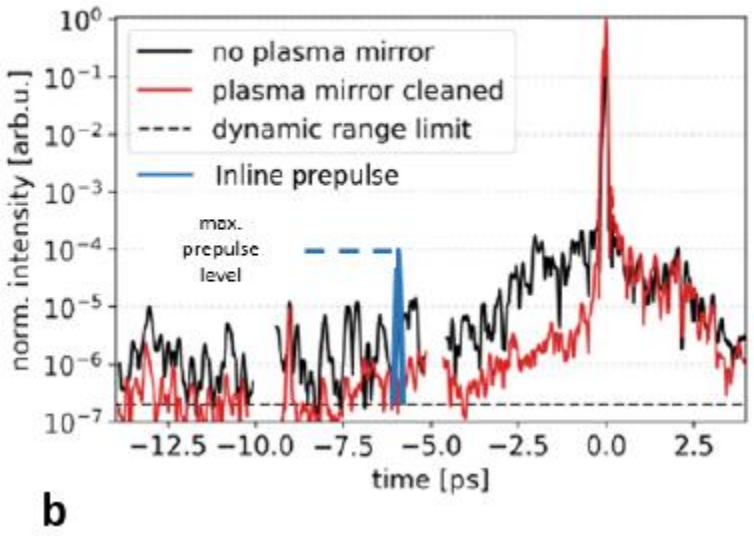
$$2\tan\beta = \frac{d_f}{l_{VPS}} = \frac{d_p}{l_{RCF} + l_{VPS} + d_{target}} \approx \frac{d_p}{l_{RCF}}$$

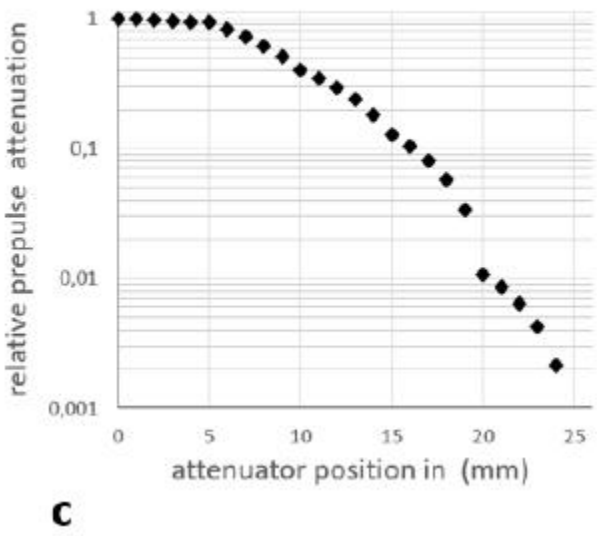
Dispu

$$\frac{\lambda_f}{l_{VPS} + d_{target} + L_P} = \frac{\lambda_f}{(6 + 0.12 + 0.74)\,\mu m} = \frac{\lambda_p}{l_{RCF}} = \frac{2.6\,mm}{56\,mm} \longrightarrow \lambda_f \approx (0.32 \pm 0.05)\,\mu m.$$

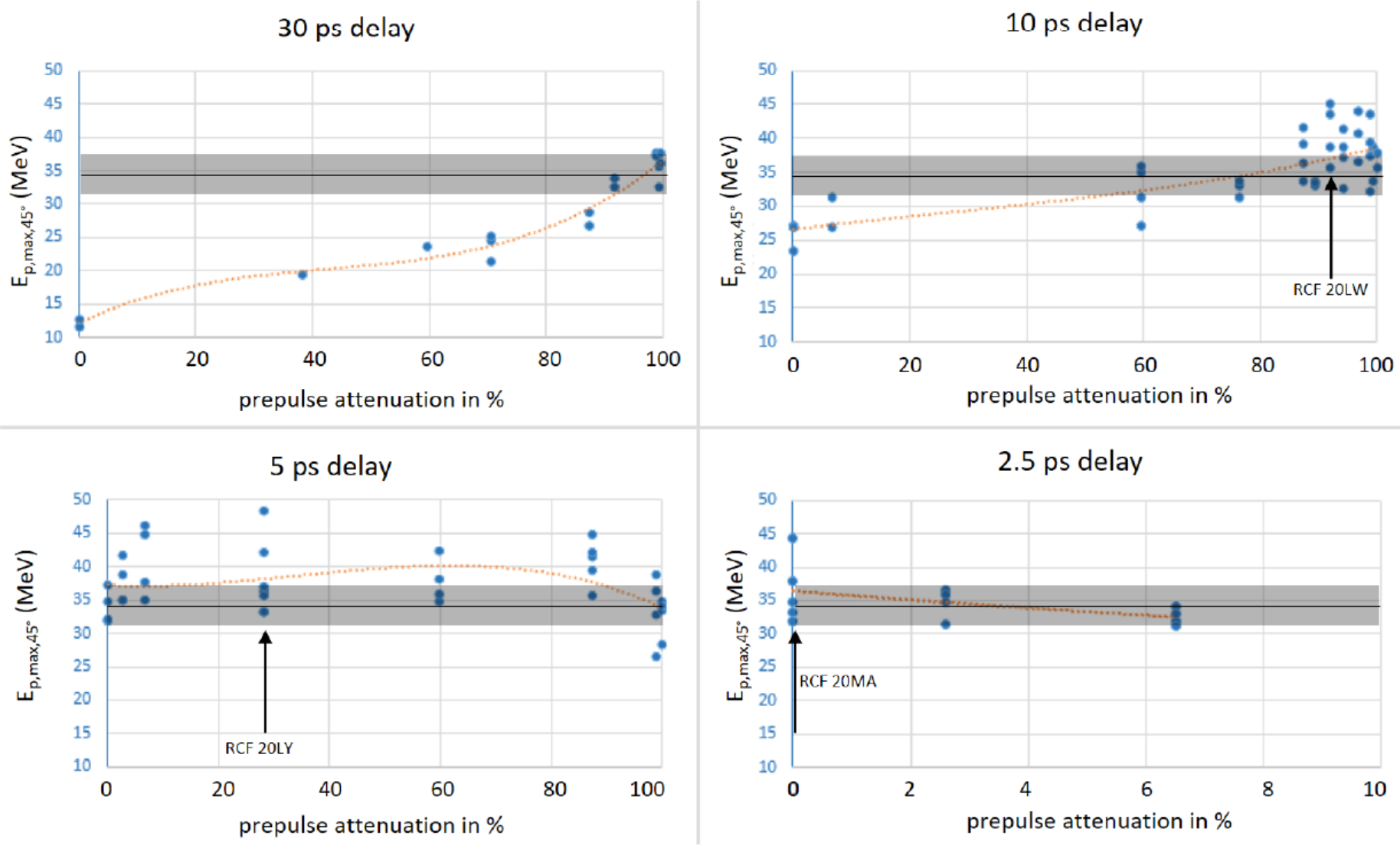
Prepulse Experiments - Setup





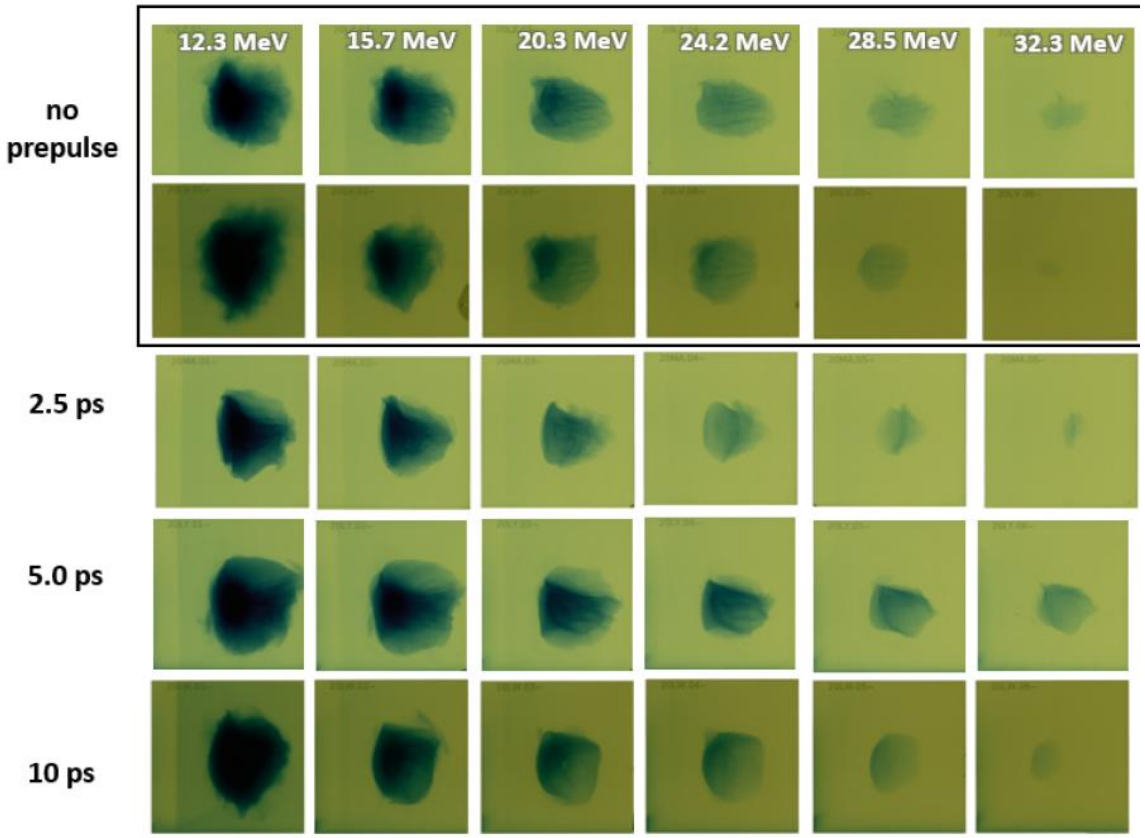


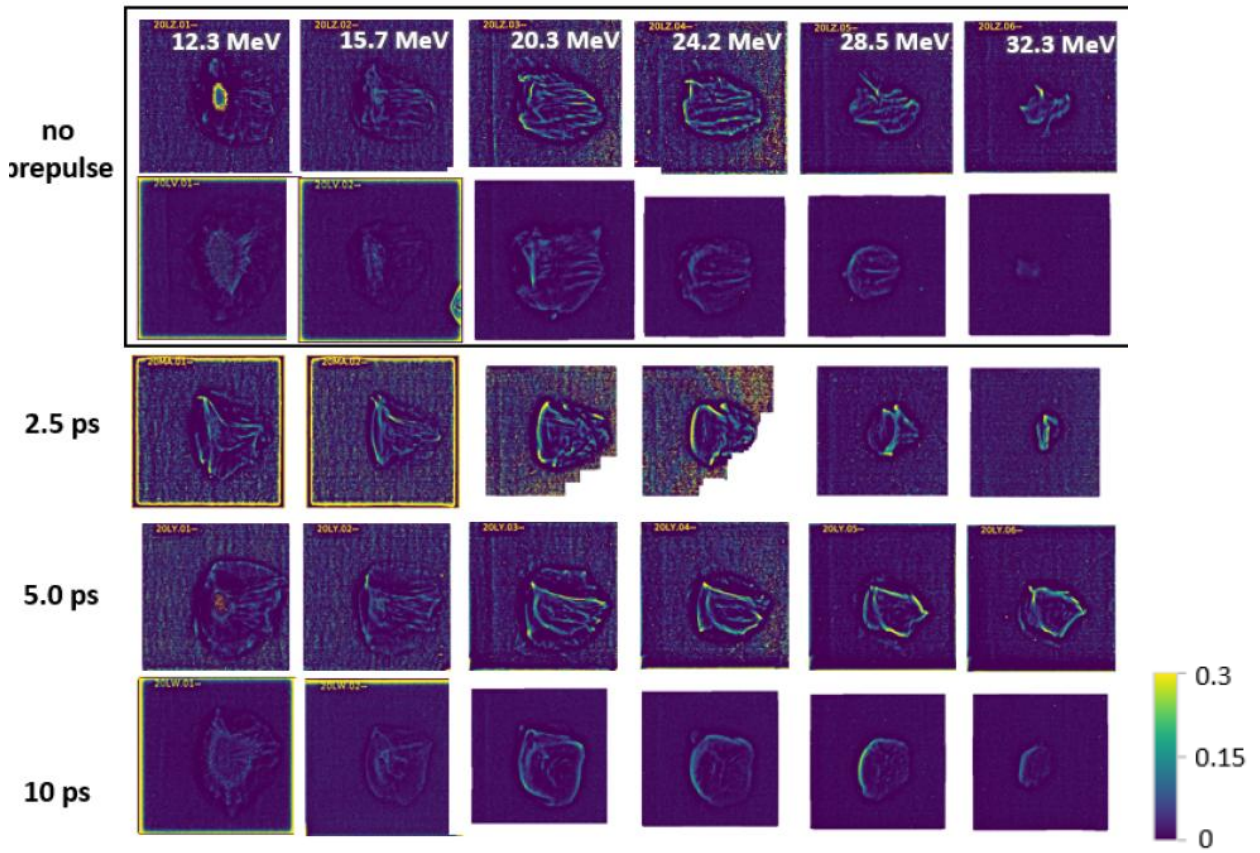
Prepulse Experiments- Proton acceleration performance



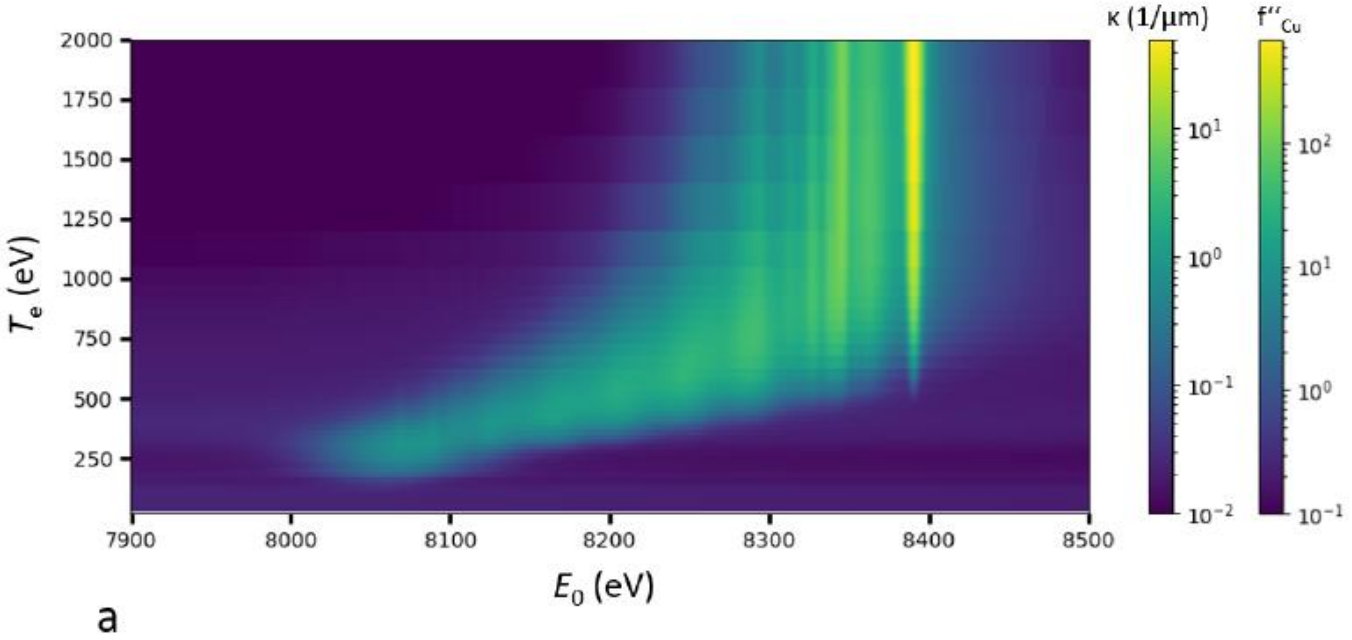
- 2µm Titan and controlled prepulse application

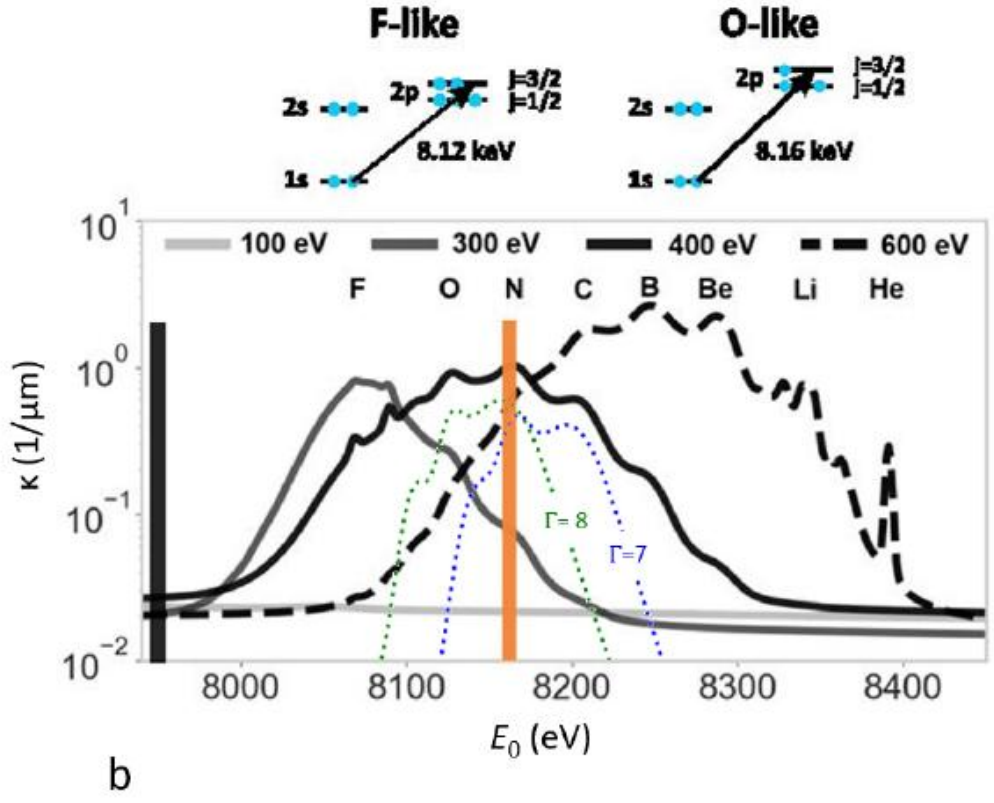
Prepulse experiments with 2µm Titanium



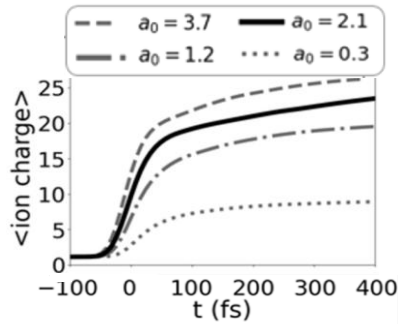


Plasma opacity obtained with FLYCHK simulations

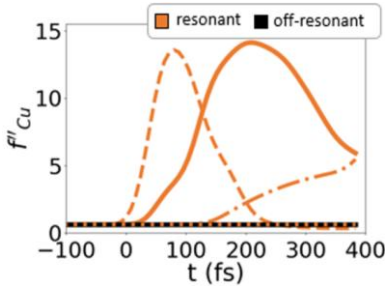


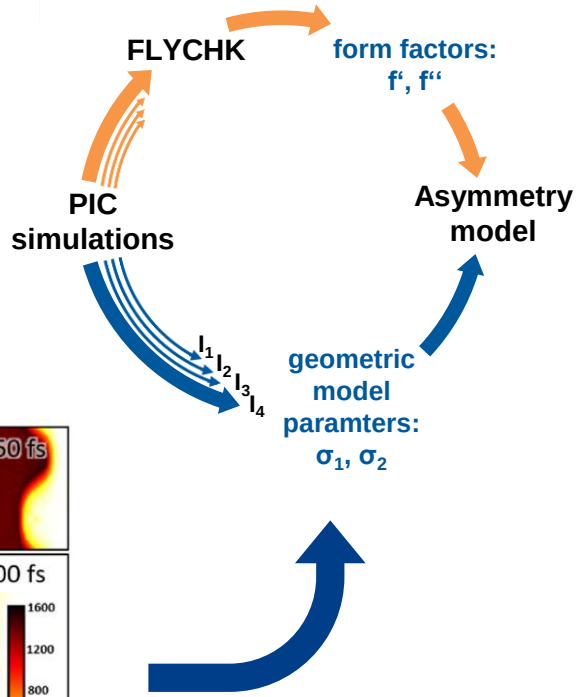


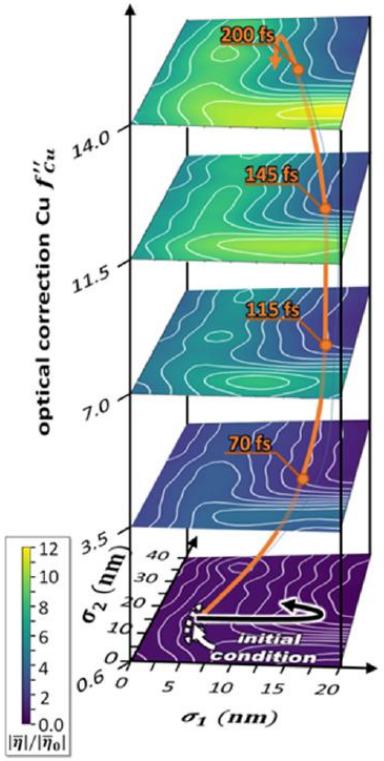
Simulations for asymmetry prediction



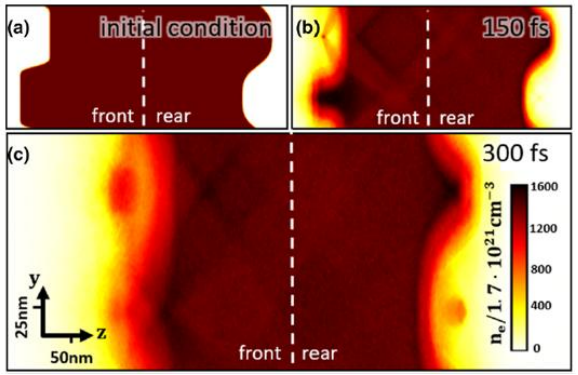
$T_e \triangleq \langle ion\ charge \rangle$

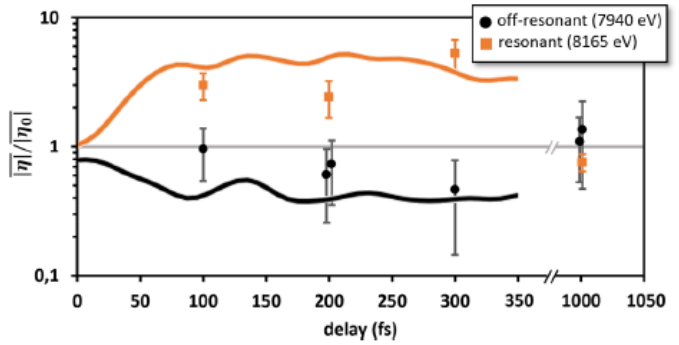




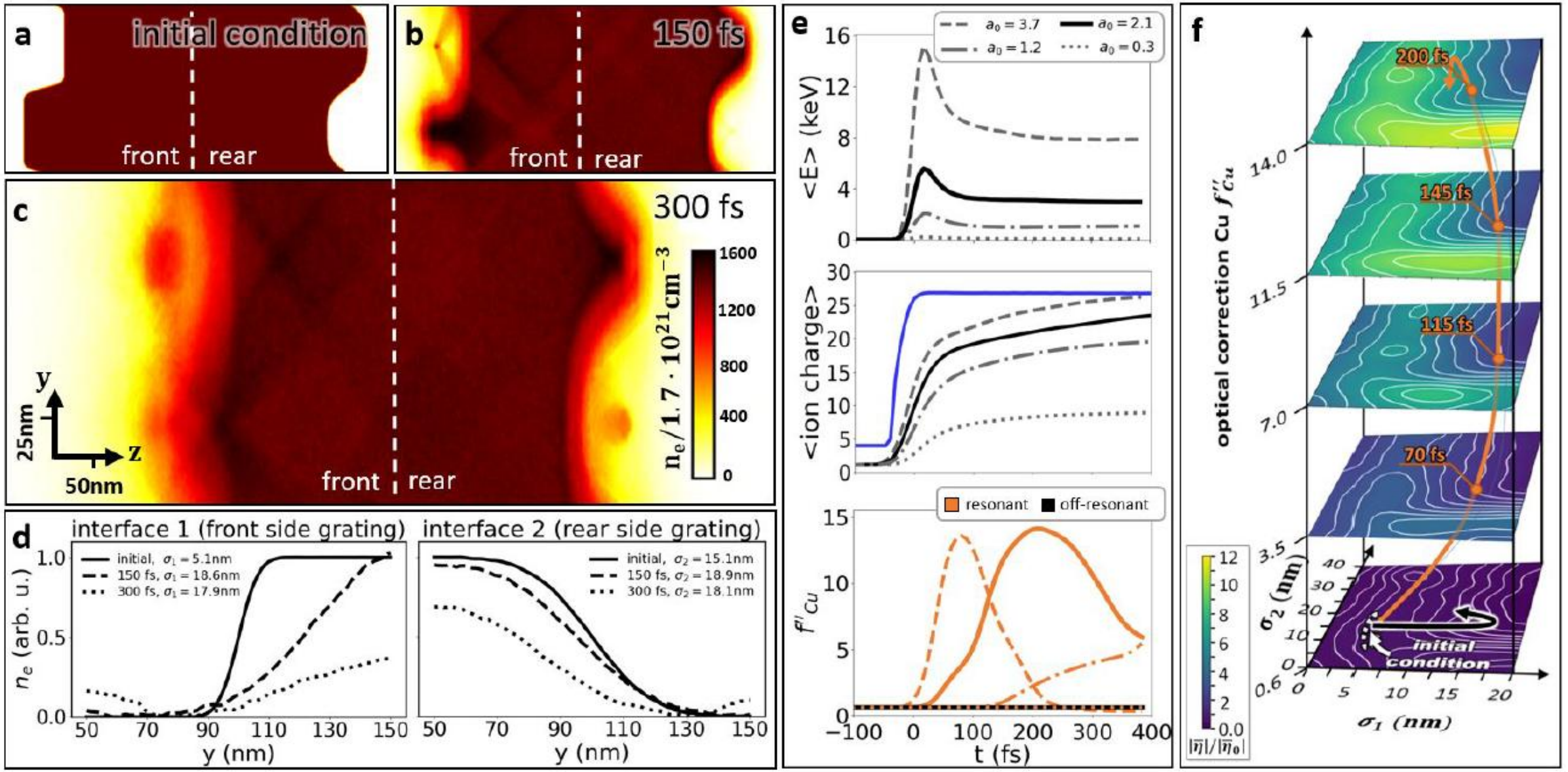


Plasma density





PIC simulations for asymmetry prediction

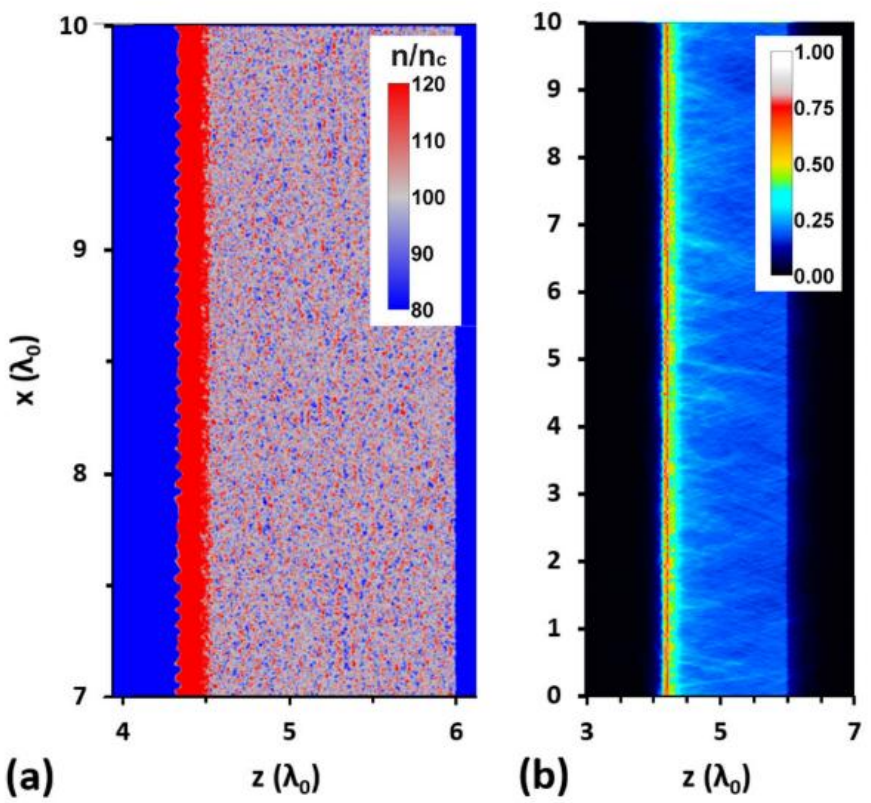


SAXS probing of complex laser-induced plasma dynamics

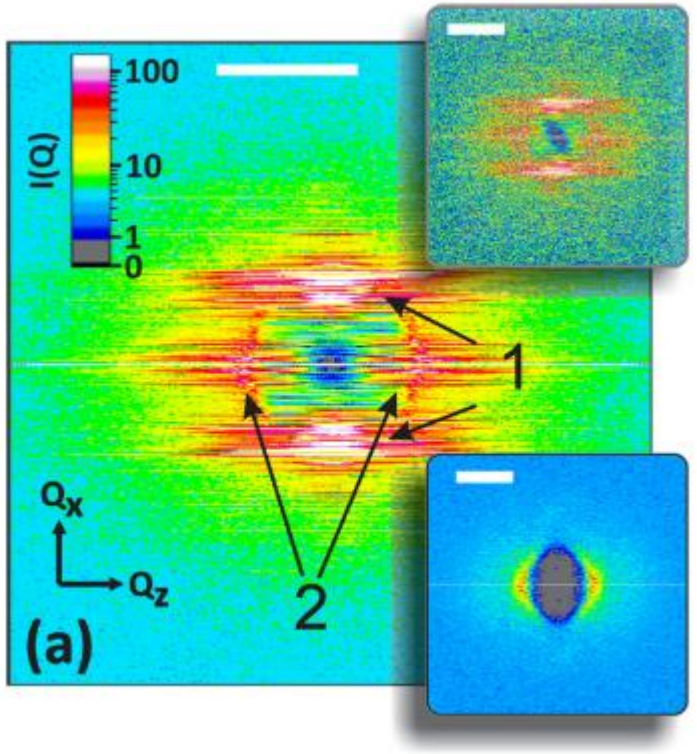
Electron density

Laser-cycle avg. electron energy density

Synthetic SAXS





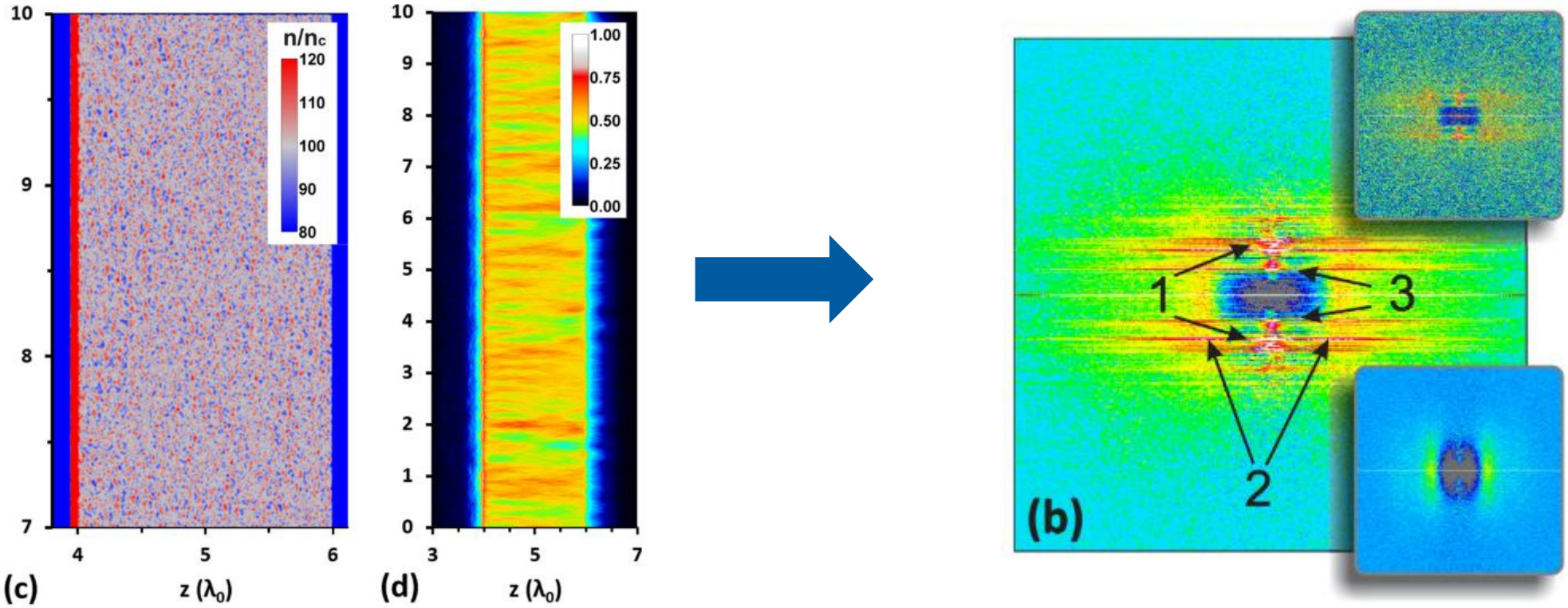


SAXS probing of complex laser-induced plasma dynamics

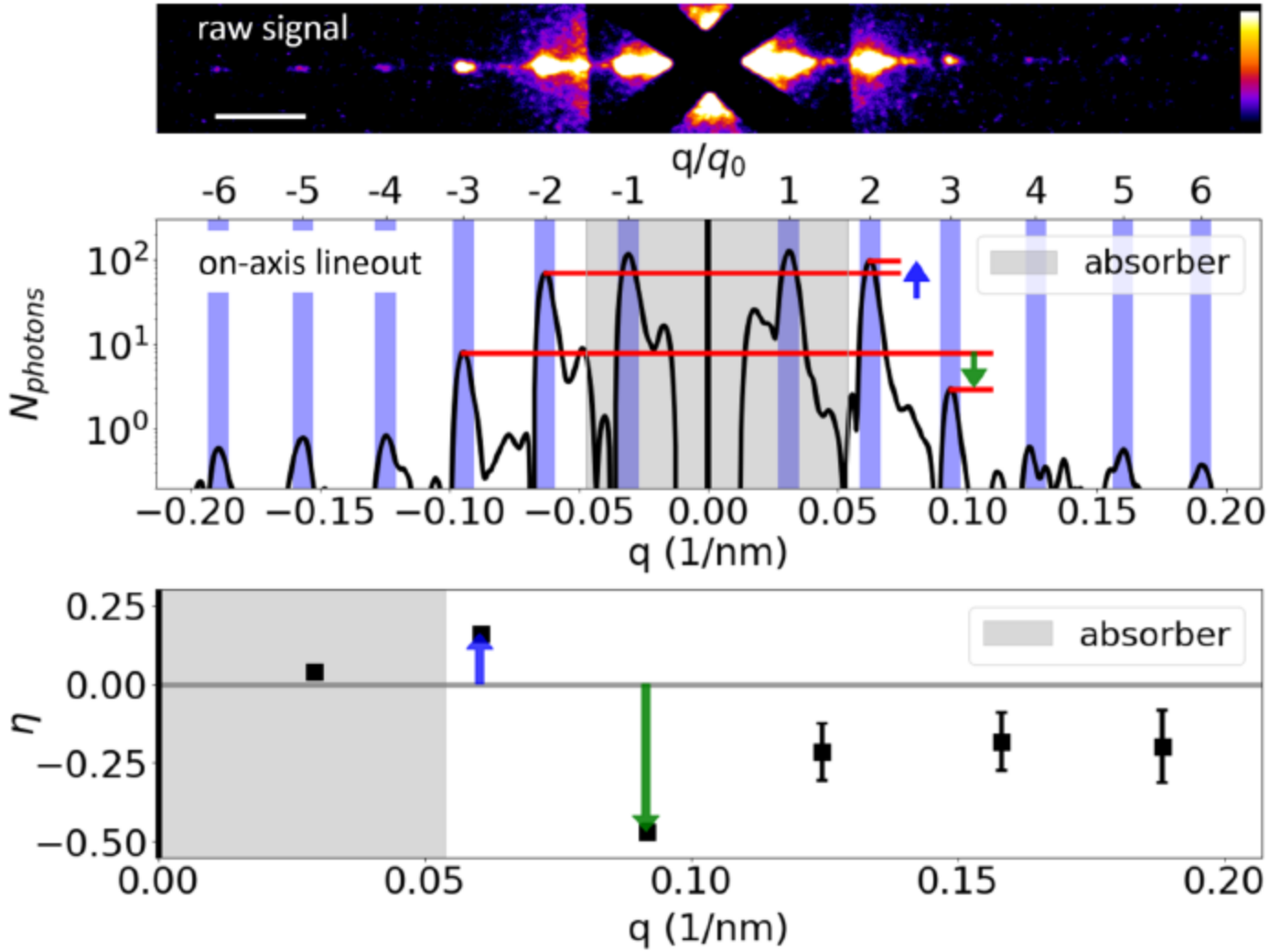
Electron density

Laser-cycle avg. electron energy density

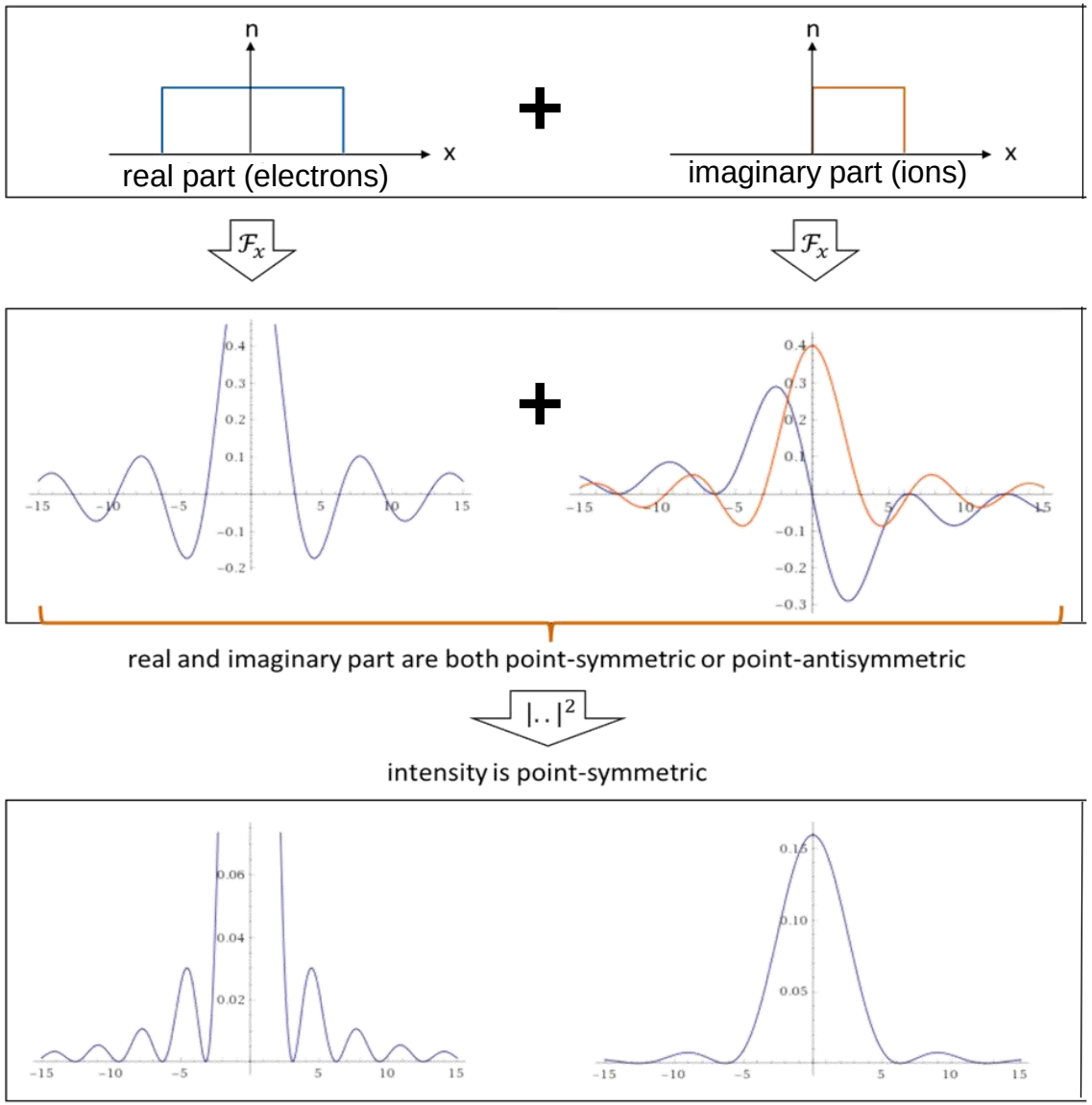
Synthetic SAXS



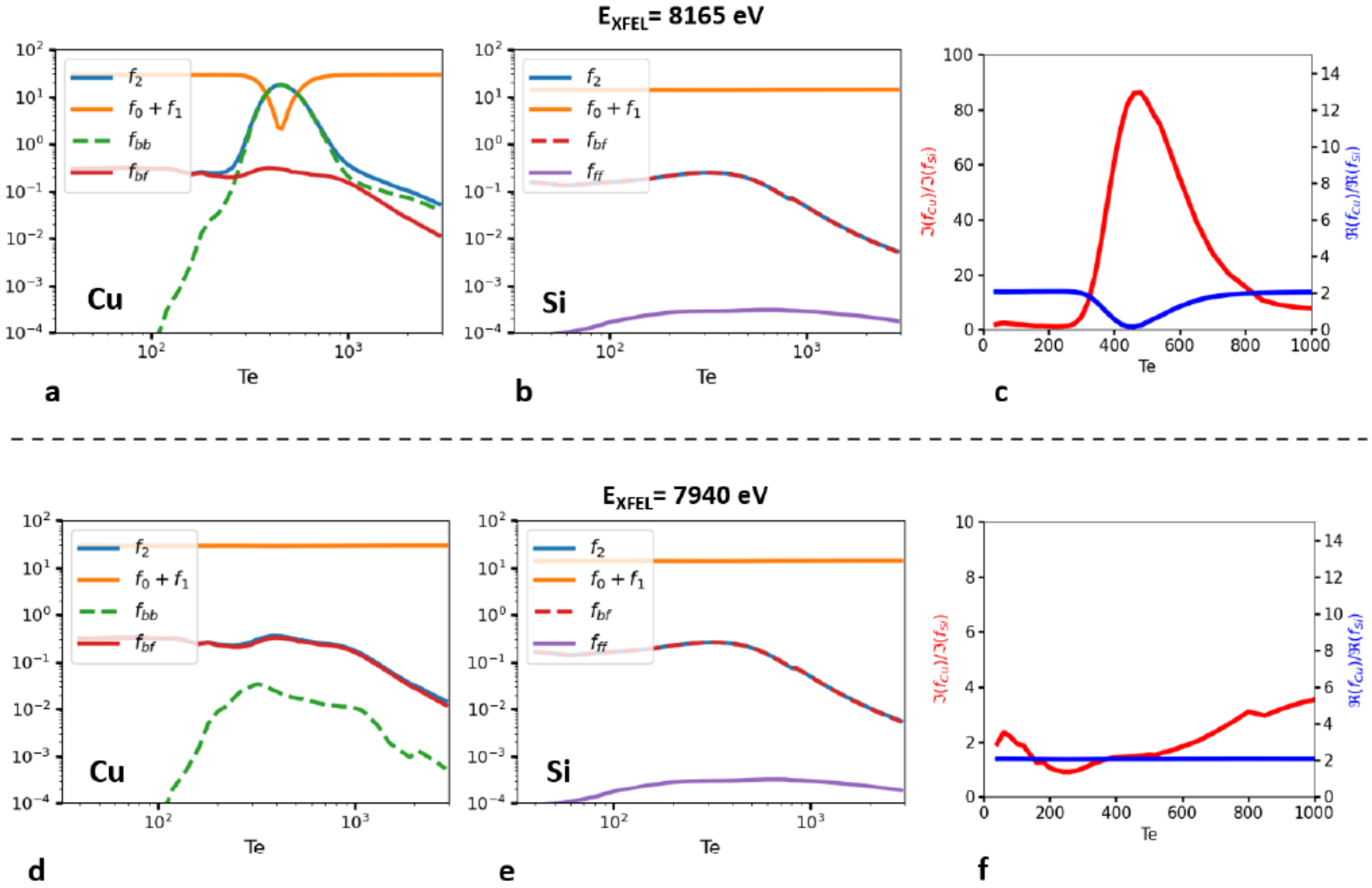
From raw SAXS data to asymmetry



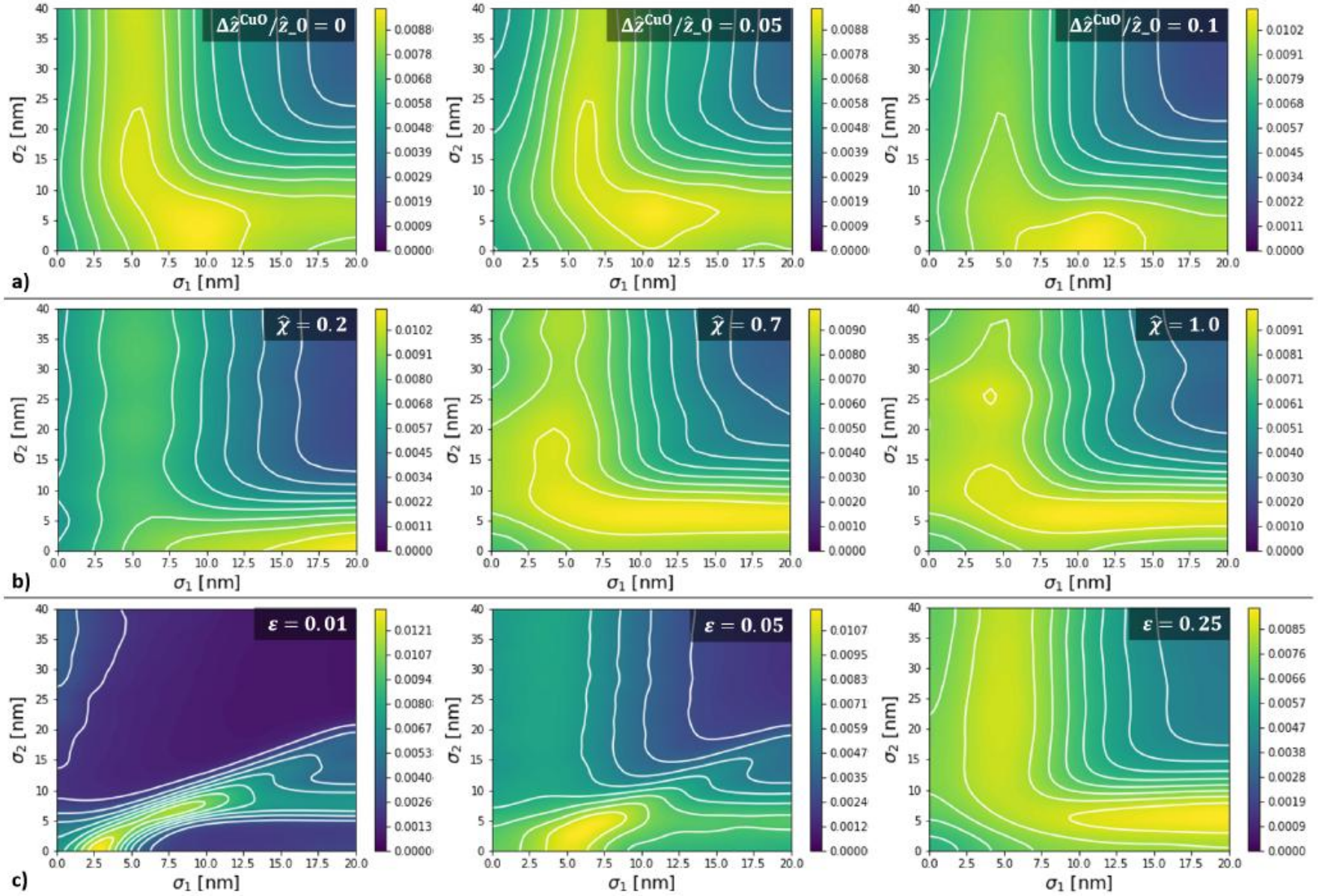
Mathematical illustration of asymmetric scattering



Differences in the real and imaginary parts of the form factors of Si and Cu

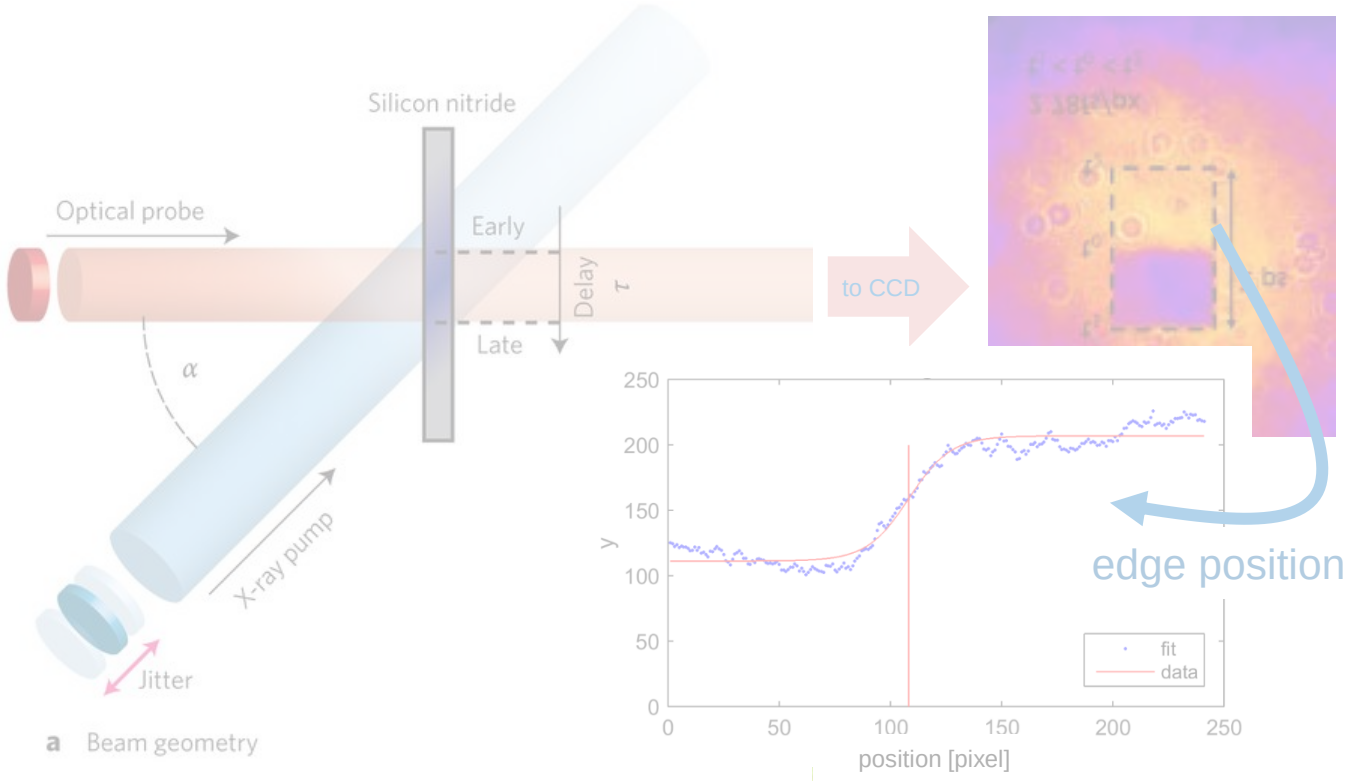


Asymmetry modeling – stability analysis by T. Kluge



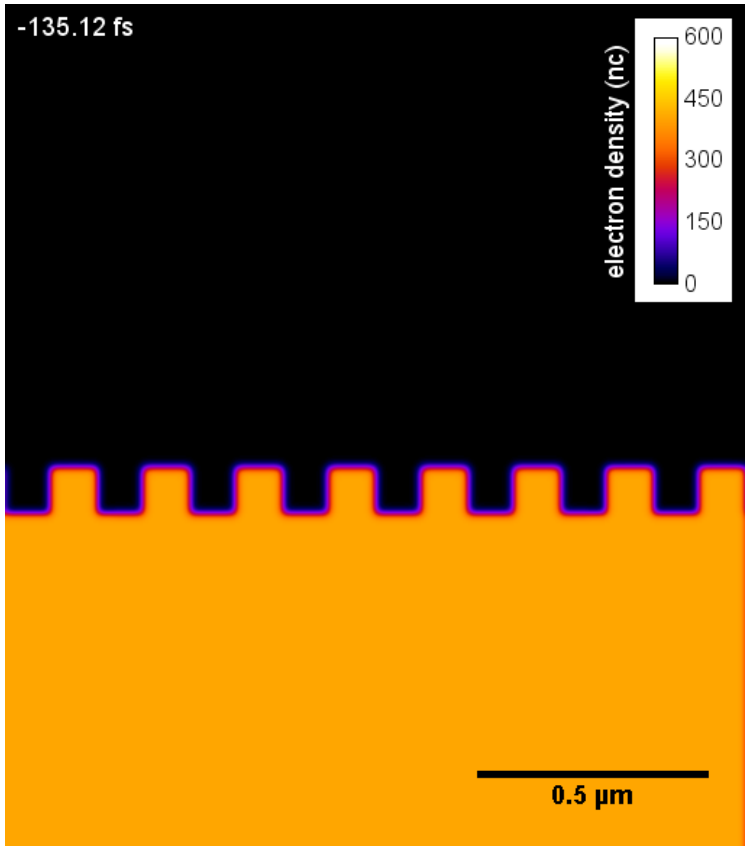
Time tool for temporal overlap

Nature Photonics 7, 215–218 (2013)
Nature Photonics 8, 706–709 (2014)

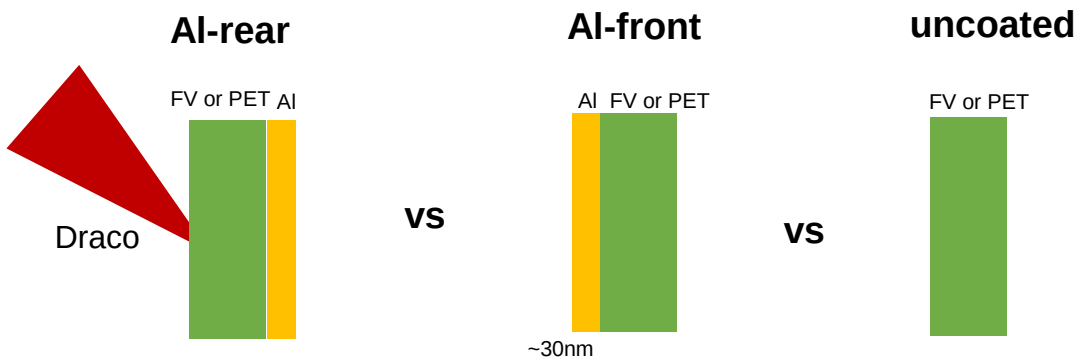


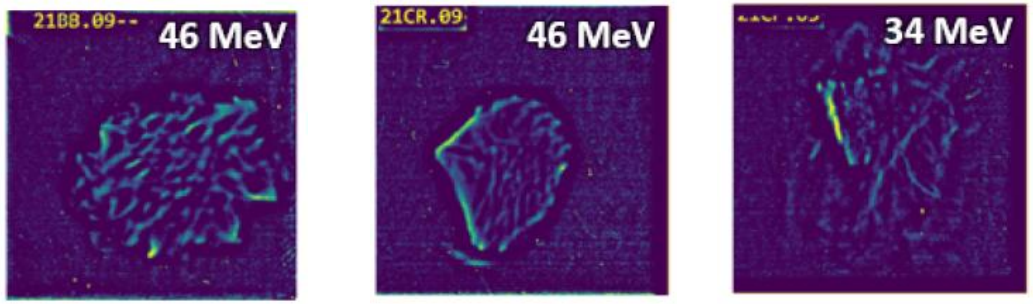
http://www.nature.com/nphoton/journal/v8/n9/fig_tab/nphoton.2014.164_F1.html

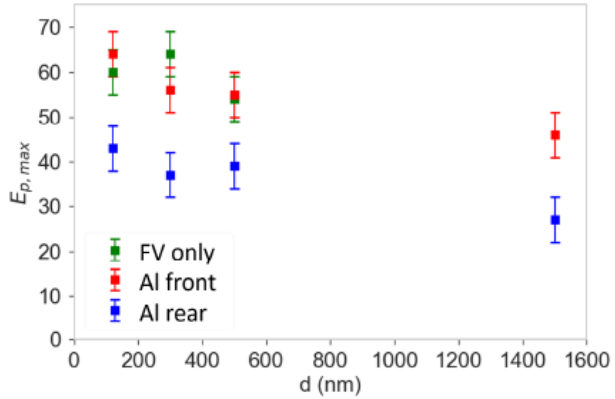
PIC simulation of the front surface expansion (LN04



Aluminium-coated plastic targets





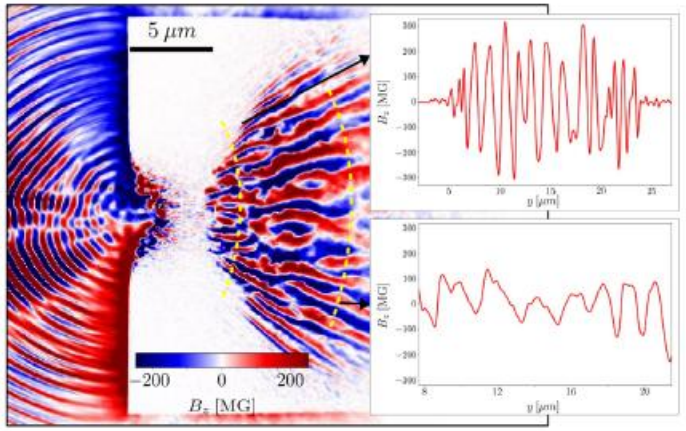


Observations:

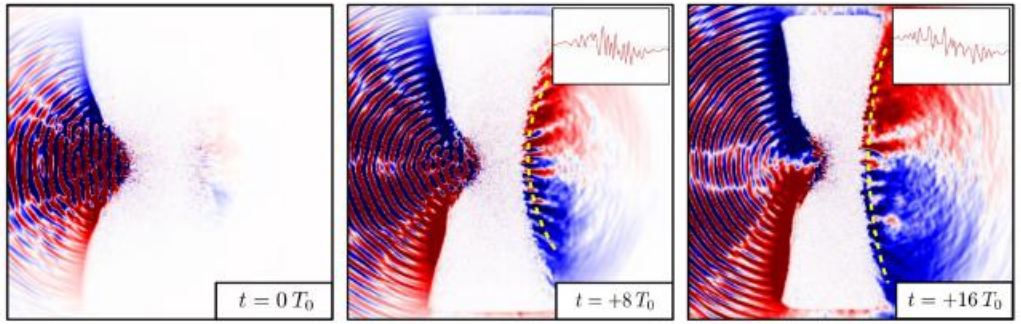
- Al at target front side: similar acceleration performance
- Al at rear: acceleration performance breaks down

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Results of target thickness and laser energy scan



a



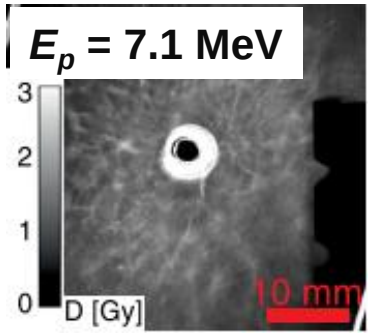
b

Filamented magnetic field structures, adapted from J. Branco

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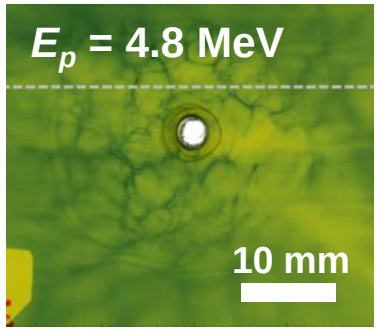
A plethora of mechanisms in the literature!

Metzkes et al.



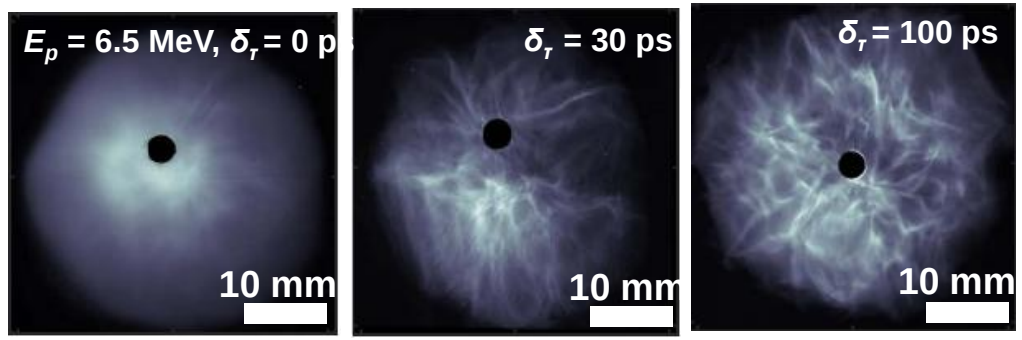
- few µm Ti,
- RTI or Weibel-like instability
- Seeded at front side

Göde et al.



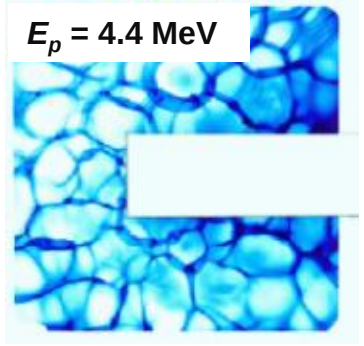
- 10 µm H_2
- Weibel-like instability
- Seeded in rear scale length

Scott et al.



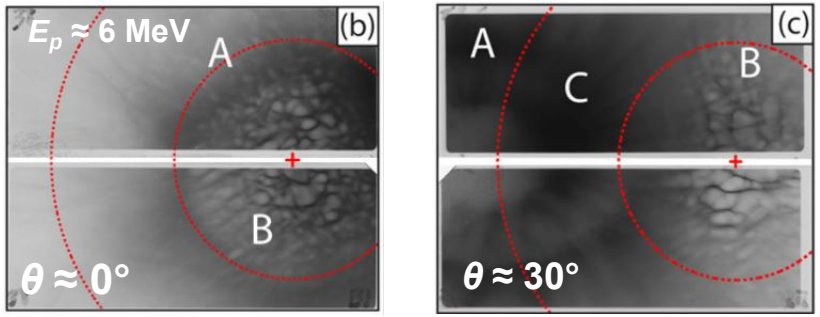
- 50 µm graphite,
- Weibel-like instability
- Seeded in rear scale length

Palmer et al.



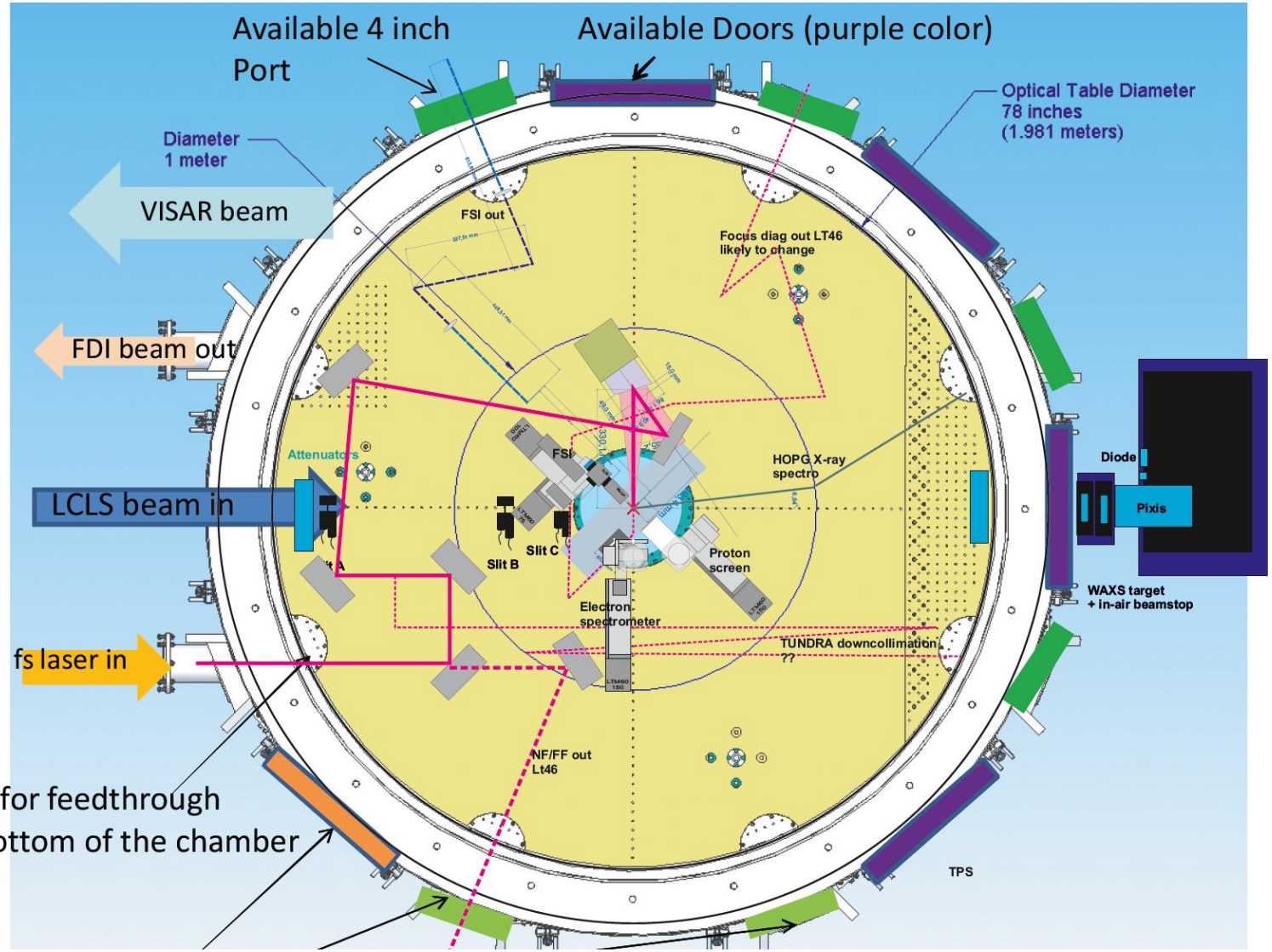
- 5 nm DLC
- RPA + RTI
- Seeded at front side

Powell et al.



- 10 nm Al
- RPA + RTI
- Seeded at front side

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